Chapter 4:
The Market Forces of Demand and Supply

Learning Objectives

4.1 explain what a competitive market is.

4.2 determine the demand for a good in a competitive market and how to apply it.

4.3 determine the supply of a good in a competitive market and how to apply it.

4.4 analyze how supply and demand together set the price of a good and the quantity sold and analyze market changes.

4.5. explain the key role of prices in allocating scarce resources in market economies.

Key Points

- Economists use the model of supply and demand to analyze competitive markets. In such markets, there are many buyers and sellers, each of whom has little or no influence on the market price.
- The demand curve for a good shows how the quantity demanded depends on the price. According to the law of demand, as the price of a good falls, the quantity demanded rises. That’s why the demand curve slopes downward.
- In addition to price, other determinants of how much consumers want to buy include income, the prices of substitutes and complements, tastes, expectations, and the number of buyers. If one of these factors changes, the demand curve shifts.
- The supply curve for a good shows how the quantity supplied depends on the price. According to the law of supply, as a good’s price rises, the quantity supplied rises. That’s why the supply curve slopes upward.
- In addition to price, other determinants of how much producers want to sell include input prices, technology, expectations, and the number of sellers. If one of these factors changes, the supply curve shifts.

Key Points

- The intersection of the supply and demand curves determines the market equilibrium. At the equilibrium price, the quantity demanded equals the quantity supplied.
- The behavior of buyers and sellers naturally drives markets toward equilibrium. When the market price is above the equilibrium price, there is a surplus of the good, which causes the market price to fall. When the market price is below

Markets and Competition

- A **market** is a group of buyers and sellers of a particular product.
- A **competitive market** is one with many buyers and sellers, each has a negligible effect on price.
- In a **perfectly competitive** market: All goods exactly the same
- Buyers & sellers so numerous that no one can affect market price—each is a "**price taker**"

Demand

- Demand is a relationship that economist use to describe the relationship between what consumers are **willing to buy** and are **able to buy**.
- Without the willingness and ability to buy a product, a consumer or group of consumers have no demand for the product in question.
- Economists and market analyst form demand relationships when a set of prices and corresponding quantities are paired.
- This pairing can be found in a demand schedule which is a table showing unique price and quantity pairings.

Demand Schedule

Let's examine market data for the iPhone in the following demand schedule:

Price	Quantity Demanded
$699	0
599	10
499	20
399	30
299	40
199	50
99	60

- Notice at a price of $699 quantity demanded is 0. There may be consumers who are willing to pay for the good but are not able. Or there may be consumers who are able to pay for the good but not willing. Or those who are not willing or able.

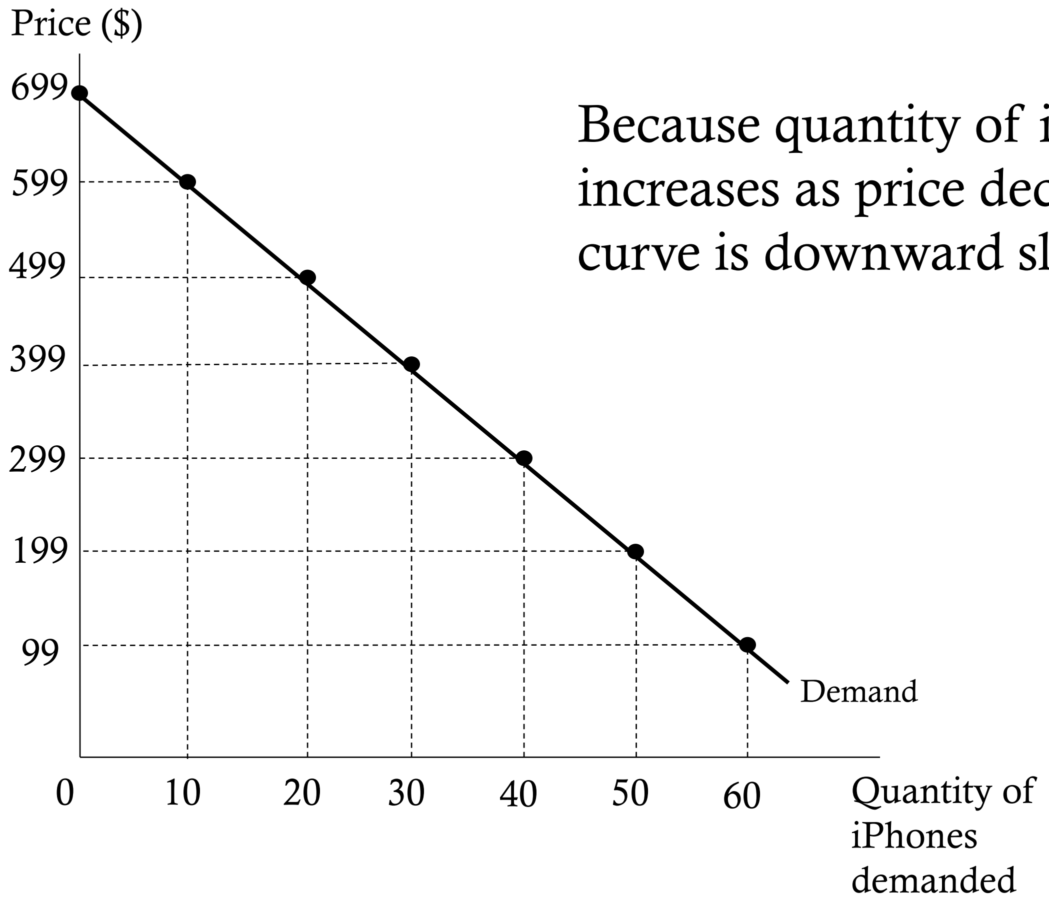
- As the price drops quantity demanded increases, up to 60 iPhones at a price of $99.

Law of Demand -There is an inverse relationship between the price of a good and the quantity demanded of the good, *ceteris paribus*.

- *Ceteris paribus* means other things held constant

The Demand Curve

Price	Quantity Demanded
$699	0
599	10
499	20
399	30
299	40
199	50
99	60

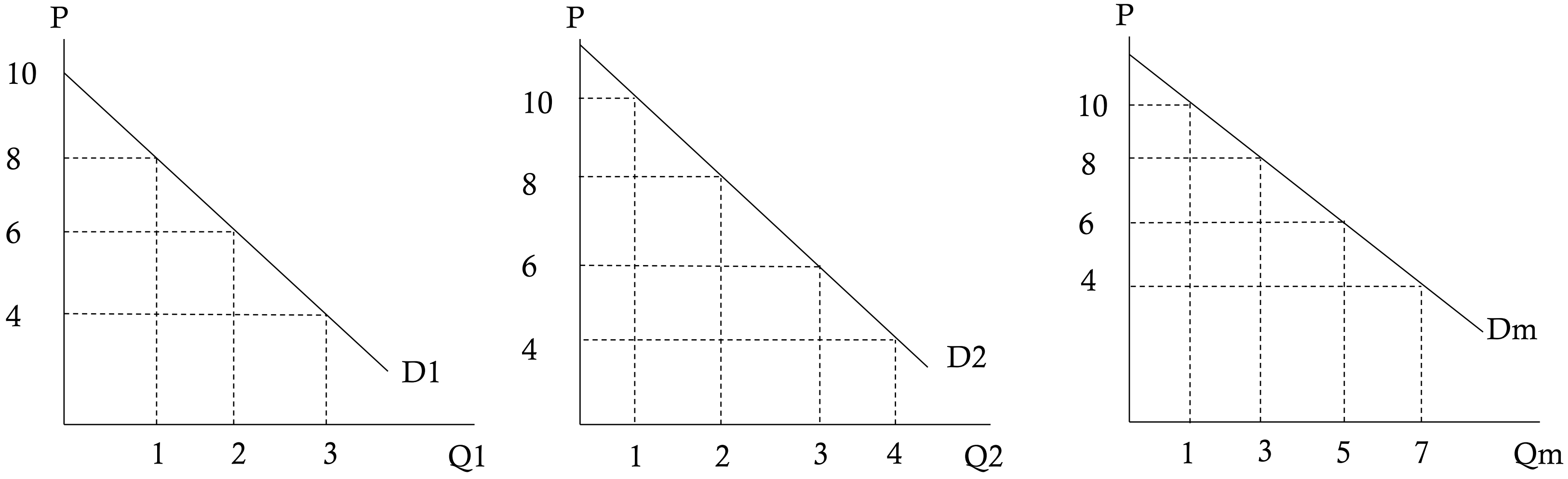


Because quantity of iPhones demand increases as price decreases the demand curve is downward sloping.

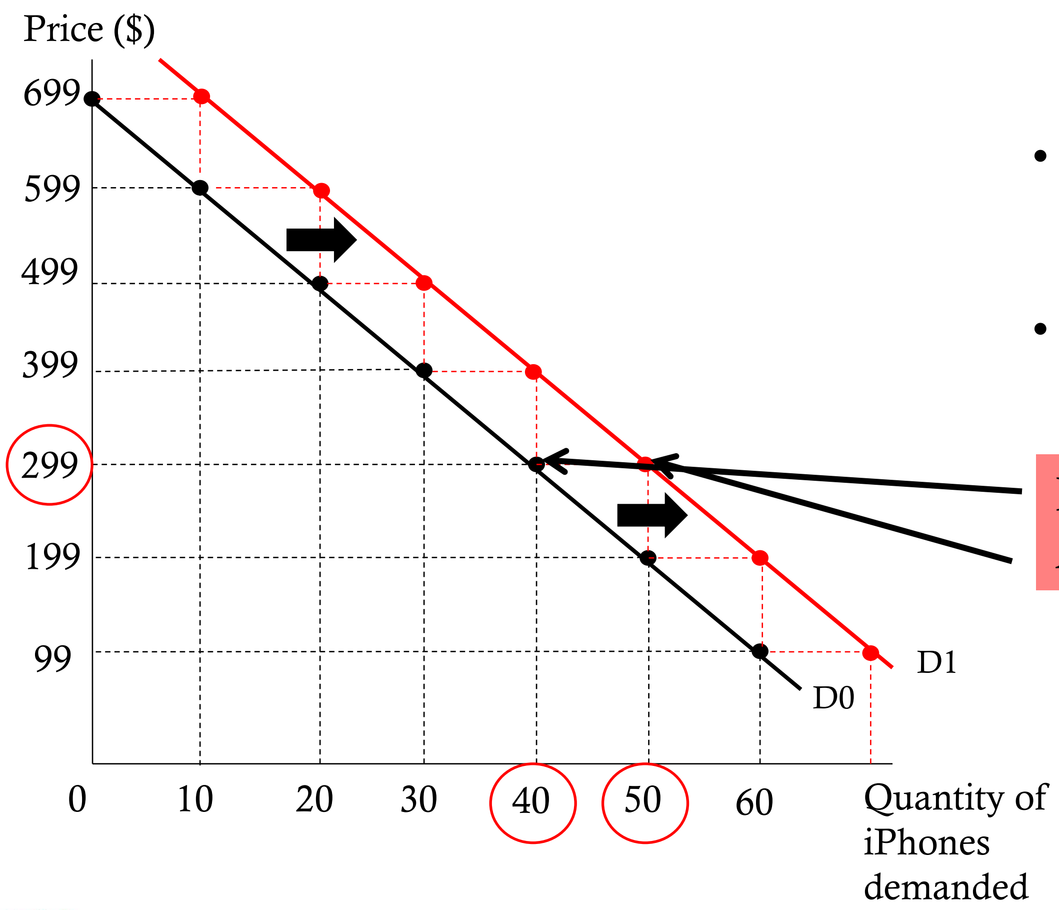
Market Demand

Price	QD1	QD2		Dm
$10	0	1		1
8	1	2	=	3
6	2	3		5
4	3	4		7

Market demand curves are formed by horizontally summing all the individual demand curves.



Shifting the Demand Curve



- A change in one of the determinants of demand causes a change in each quantity data point.
- An increase in demand will cause quantity demanded to increase at each previous price.
- Compare the quantity demanded at a price of $299

Before on demand curve D0: Q = 40
After on demand curve D1: Q = 50

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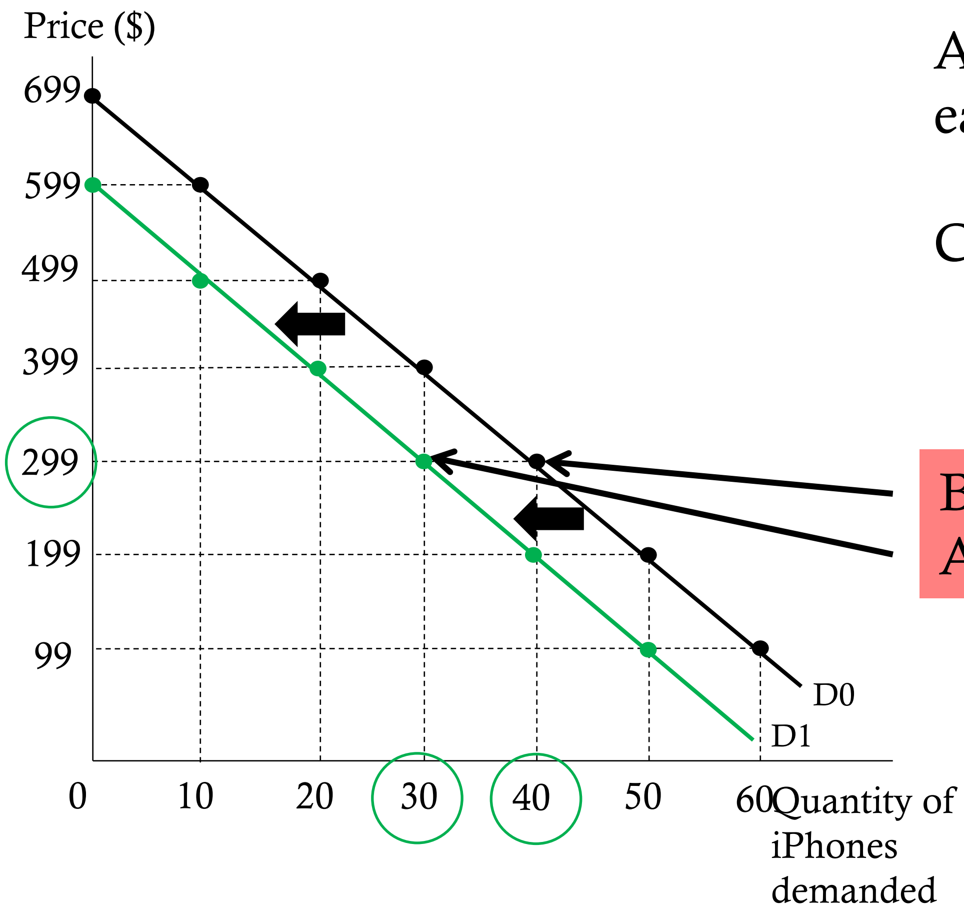
Shifting the Demand Curve



An decrease in demand will cause quantity to fall at each previous price.

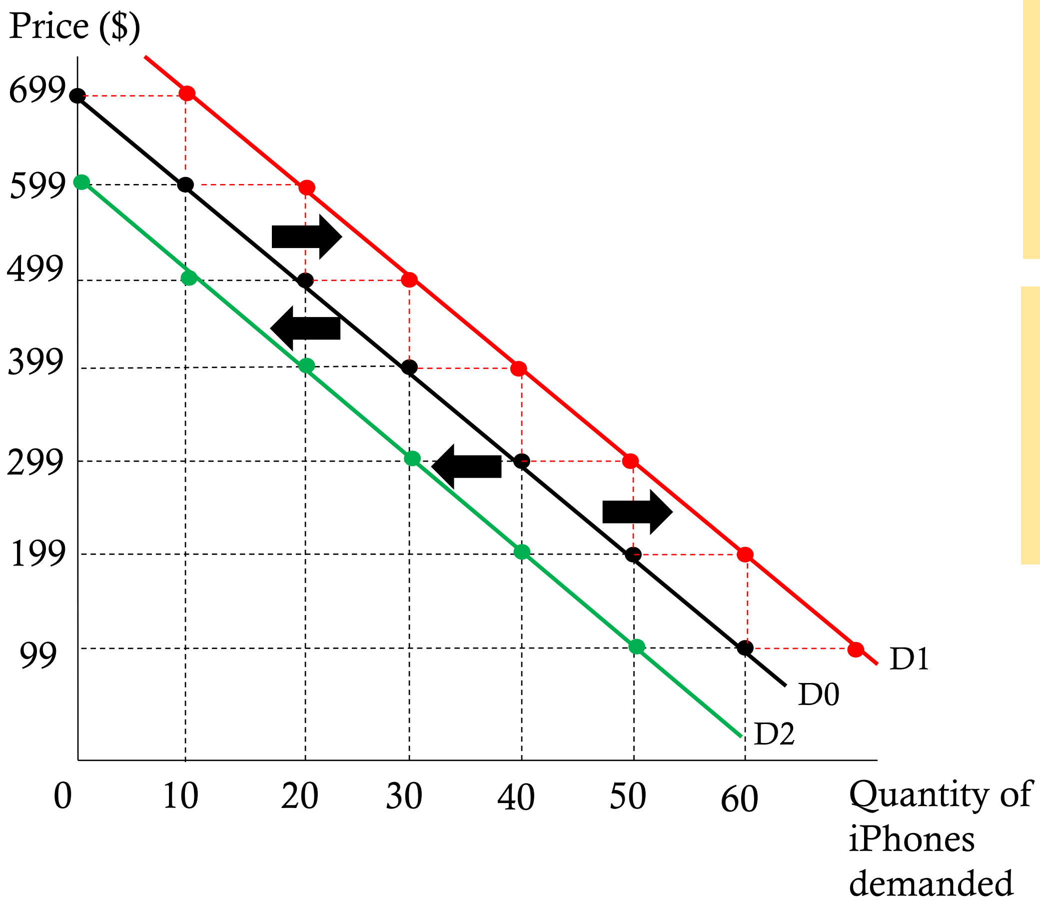
Compare the quantity demanded at a price of $299

Before on demand curve D0: Q = 40
After on demand curve D1: Q = 30

What Shifts Demand?

- The ceteris paribus assumption (all things equal/held constant)
- Remove that assumption to allow factors to affect demand
- Factors (determinants)
  - Tastes/Preferences
  - Population (size and composition)
  - Prices of Related Goods
  - Income
  - Expectations

Size/Composition and Tastes



Increases in the size/composition/taste for a product increases demand. Demand shifts from Do to D1.

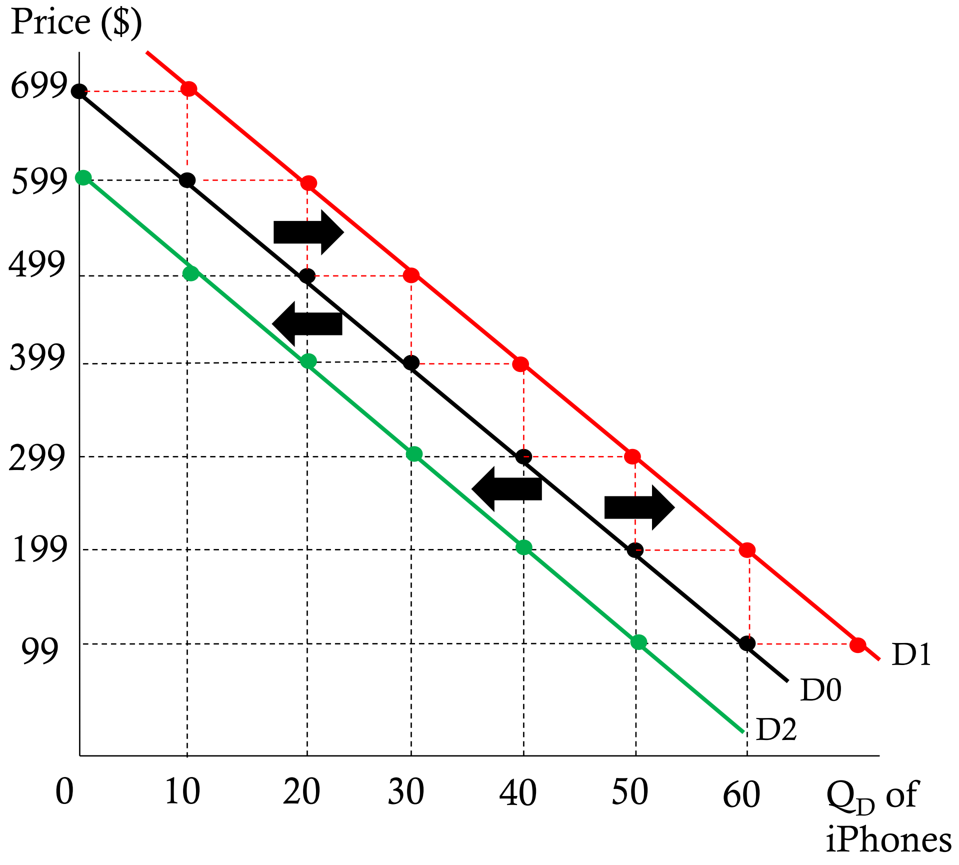
Decreases in the size/composition/taste for a product decreases demand. Demand shifts from Do to D2.

Change in Tastes – Going Vegan

- Kroger launches private-label vegan beef as demand for meat substitutes soars
- "Kroger is launching new private-label vegan ground beef and burger patties as it seeks to capitalize on growing consumer demand for plant-based meatless substitutes, the company said Wednesday.
- The nation's largest grocery store chain is among many companies creating their own versions of vegan meats that closely mimic the taste of real meat products. Meat producers like Smithfield and Hormel and Big Food companies like Kellogg are trying to jump on the trend sparked by veggie burgers made by Beyond Meat and Impossible Foods."

Prices of Related Goods

The analysis of related goods hinges on a change in the price of one good and the resulting change in the quantity demanded of the other good.



Complement good – goods that are consumed together.

An **increase** in the price of wireless service will cause a **decrease in the demand** for iPhones

Substitute good – a good that can be replaced by another with similar characteristics.

An **increase** in the price of the S3 will cause an **increase in the demand** for iPhones.

Complement		**Substitute**	
Price wireless service	↑	Price Galaxy	↑
Demand iPhone	↓	Demand iPhone	↑
Price wireless service	↓	Price Galaxy	↓
Demand iPhone	↑	Demand iPhone	↓

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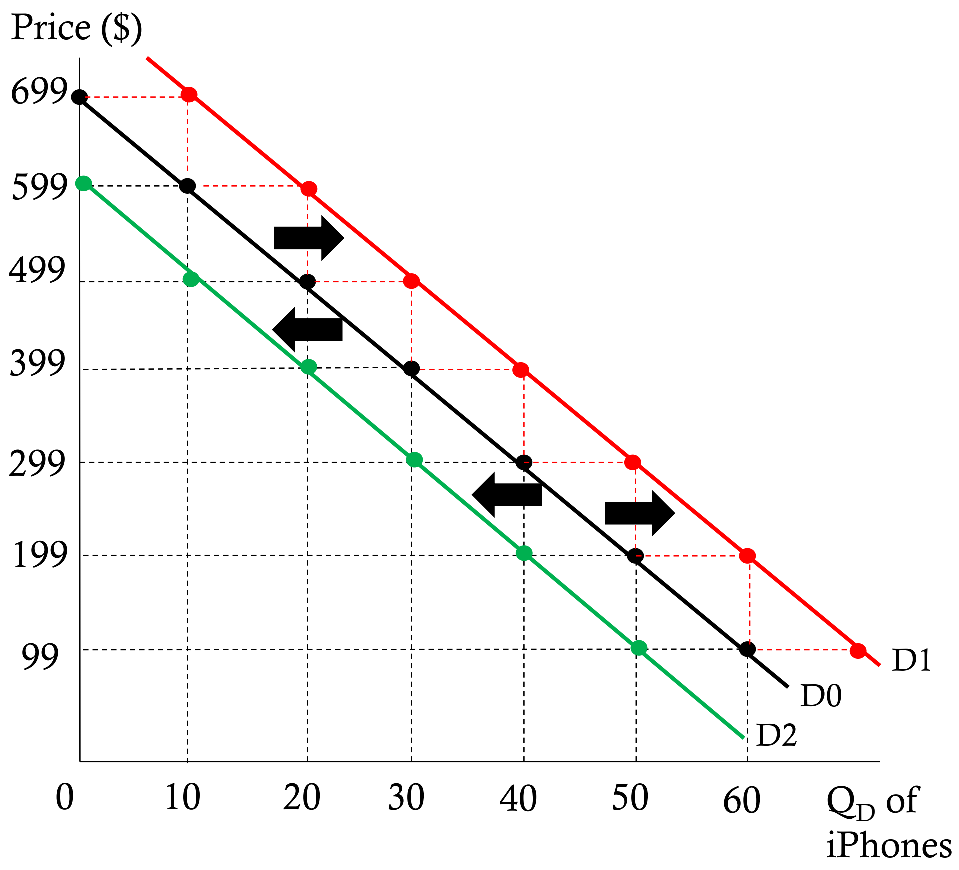
Price of Related Goods – No Longer Complementing?

- Why Walmart is seeing increased sales for tops, but not bottoms during the coronavirus crisis
- “With more and more people working from home, Walmart has picked up on an interesting trend: Tops have seen an increase in sales, while bottoms haven't.
- The reason? Teleworking.
- That's what Walmart's executive vice president of corporate affairs Dan Bartlett told Yahoo Finance on Thursday. Later, a spokesman for the company told CNN the same thing.”

Changes in Income



A change in income causes consumers to purchase more of some types of goods and less of other types of goods.

Normal Good
An **increase** in income **increases demand**

Inferior Good
A **decrease** in income **increases demand**

Normal Good
A **decrease** in income **decreases demand**

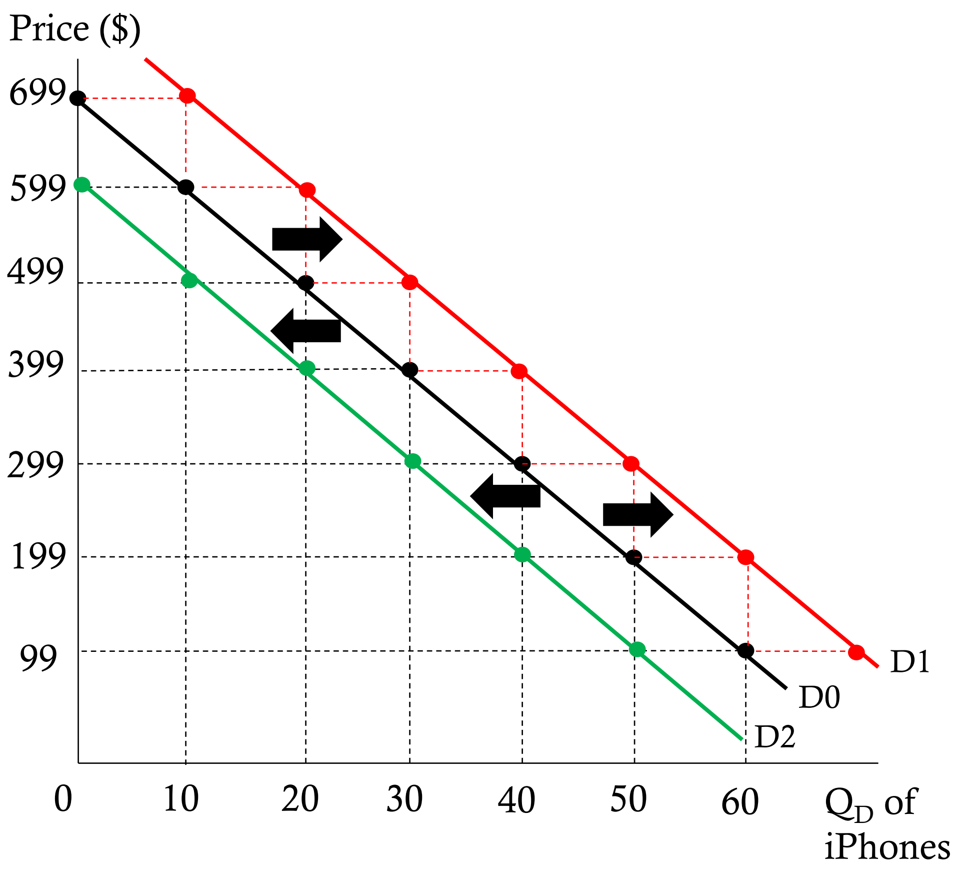
Inferior Good
An **increase** in income **decreases demand**

For the iPhone, consumers would purchase more if their income went up making it a normal good. An inferior good might be a non-smart phone.

Change in Income – Beefing Up

- How McDonald's Thrived During the Recession
- "An exception to all of this economic despair has been fast food chain **McDonald's** (MCD). While rivals **Burger King** (BKC) and **Yum! Brands** (YUM) struggled during the recession, McDonald's has not only stayed afloat since 2008: it has grown."
- "In August, McDonald's reported that sales at stores open at least 13 months rose 4.6%."
- Things that McDonald's focused on: Recession-Friendly Pricing, New Products for Different Markets, Reduced Advertising Costs. Improved Operations
- **Rapid Price Adjustments**
  - "The same *Wall Street Journal* piece also discusses how McDonald's has begun using computer systems for in-store decision making. Using the "reams of customer data" at its disposal, McDonald's now closely analyzes "everything from whether customers are trading down to smaller value meals or dropping cokes from their orders to exactly how much they're willing to pay for a Big Mac." Alvarez, who confessed to loving numbers, said that these kinds of computerized systems allow McDonald's restaurants to rapidly adjust prices based on current customer demand."

Changes in Expectations



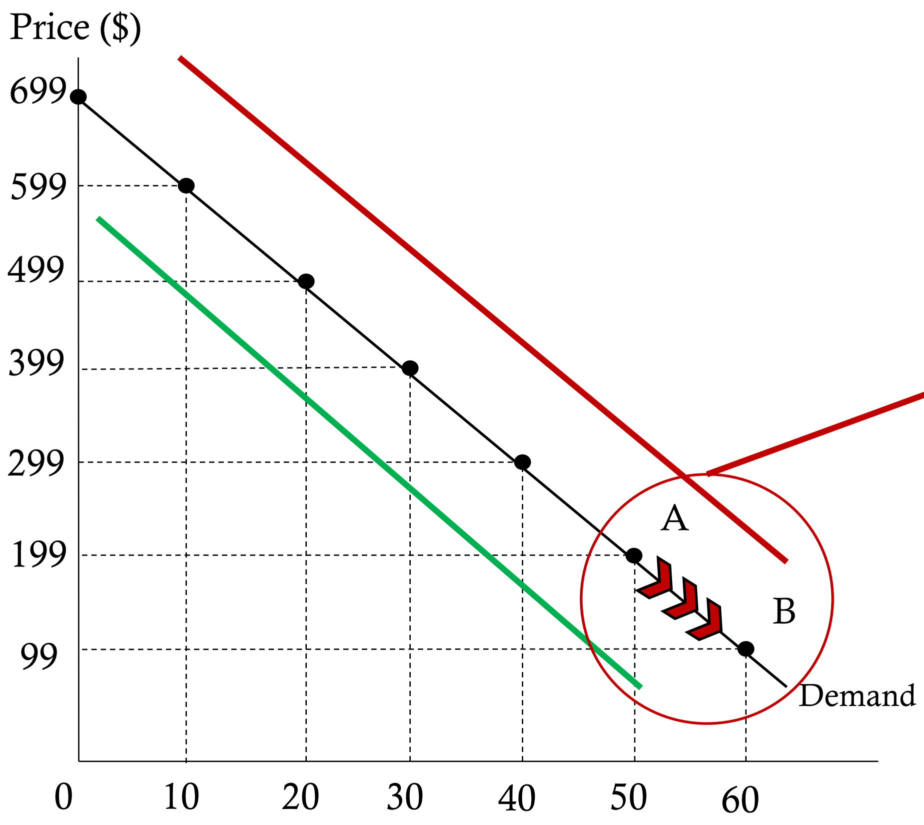
As consumers we tend to be forward looking, that is we have expectations. Our expectations affect our consumption decisions today.

Future prices
An expectation of a **price decrease** will **decrease demand** today

Future income
An expectation of an **income decrease** will **decrease demand** today

Future prices/income
An expectations of a **price/income increase** will **increase demand** today

What Causes Shifts vs. Movements



- A change in quantity demanded is represented by a movement along the demand curve (point A to B).
- If the movement is down the demand curve, price decreases and quantity demanded increases. If the movement is up the demand curve price increases and quantity demanded decreases
- A change in demand is represented by a shift in the demand curve. The previous prices remain, and quantity will increase or decrease

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Supply

- Supply is the relationship between the price of a good and the amount of a good the producer is willing and able to supply
- Like demand, there has to be a willingness and ability to sell the product, without it there will be no supply.
- Quantity supplied comes from a production process where the factors of production (land, labor, capital and entrepreneurial ability) are combined to create a product to be sold in the market.
- The unique pairing of quantity supplied and price can be put into a table or supply schedule.

Supply Schedule

Below is the supply schedule for the iPhone:

Price	Quantity Supplied
$699	60
599	50
499	40
399	30
299	20
199	10
99	0

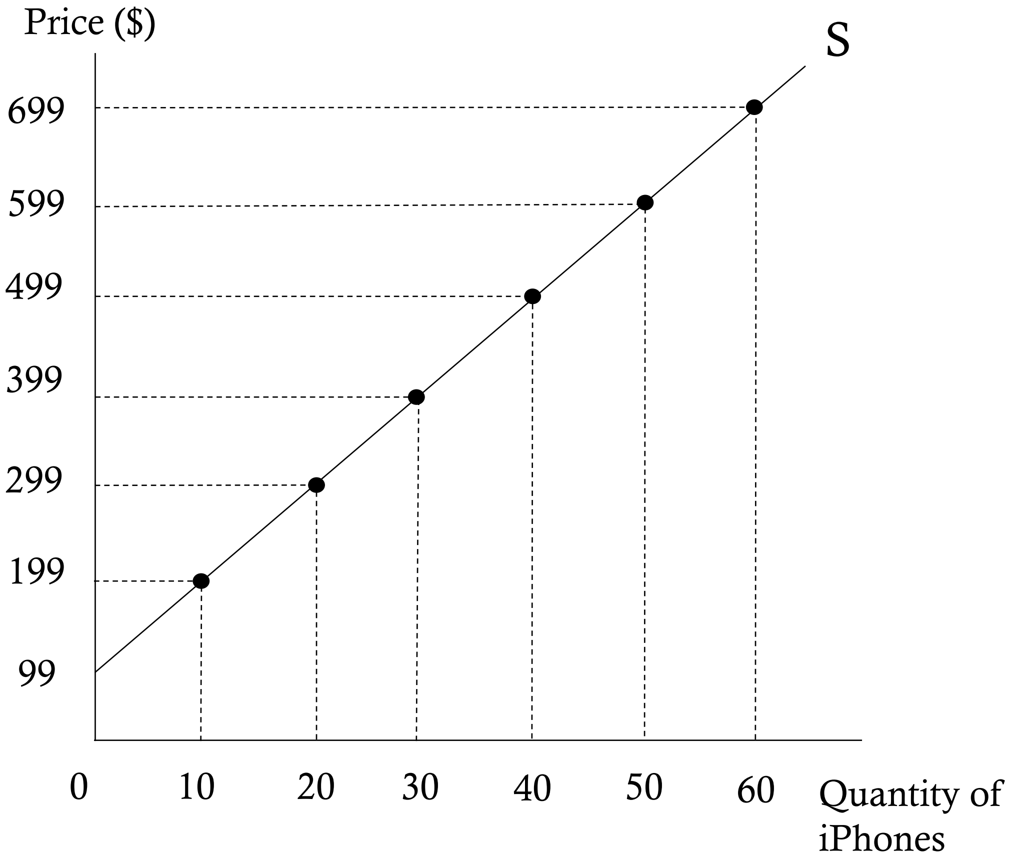
At a price of $699, the producer is willing and able to sell 60 iPhones.

As the market price falls from $699 to $99, quantity supplied does as well (from 60 to 0).

Law of Supply – there is a direct relationship between quantity supplied and price, *ceteris paribus.*

The Supply Curve

Price	Quantity Supplied
$699	60
599	50
499	40
399	30
299	20
199	10
99	0



Each price and quantity point are matched on their respective axis. Because quantity increases as price increases, supply is positively sloped

Market Supply

Market supply, like market demand, is the horizontal summation of all firm supply curves.

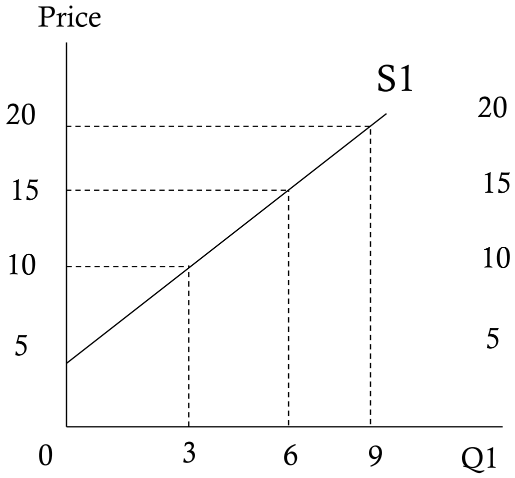
Price	Quantity
5	0
10	3
15	6
20	9

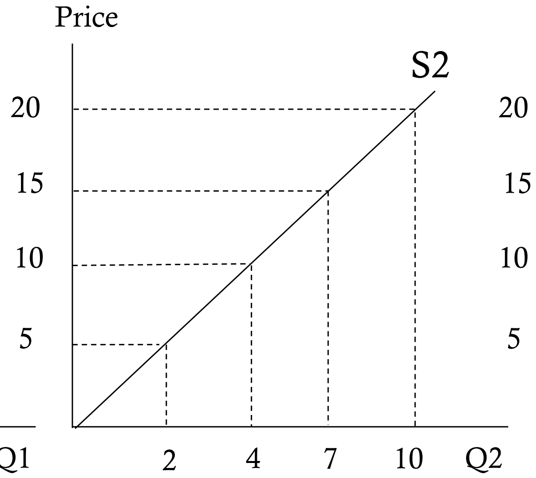
\+

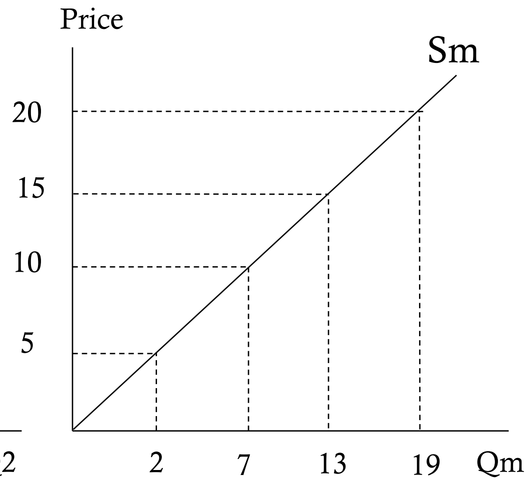
Price	Quantity
5	2
10	4
15	7
20	10

=

Price	Quantity
5	2
10	7
15	13
20	19







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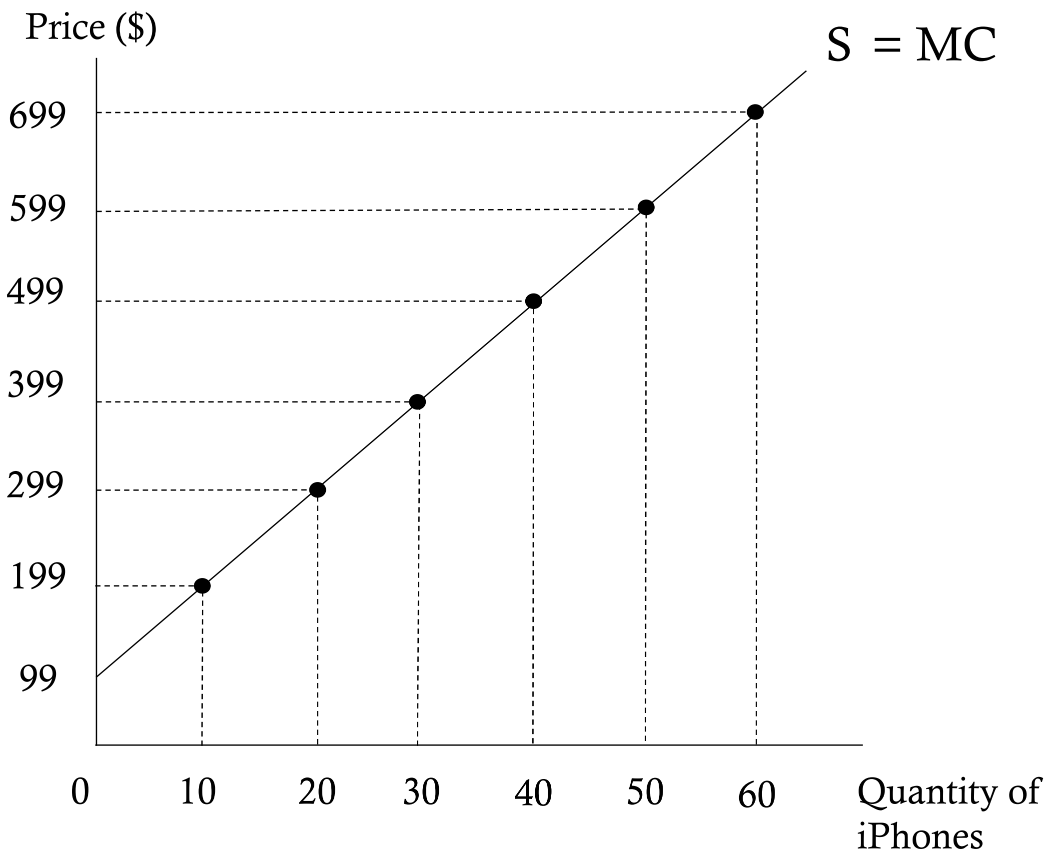
The Supply Curve

- Supply curves are governed by the marginal cost of production.
- The firm will only sell an additional unit of output if the price compensates the firm for doing so.
- The supply curve therefore can be viewed as the marginal cost curve.



Shifting the Supply Curve

- An increase in the supply curve causes the entire curve to shift to the right
- At the each of the previous prices, quantity supplied has increased from its original values.
- For example, at a price of $399

Before
Quantity supplied = 30

After
Quantity supplied = 40

Shifting the Supply Curve

- A decrease in the supply curve causes the entire curve to shift to the left.
- At each previous price, quantity supplied has been reduced.
- For example, at a price of $499:

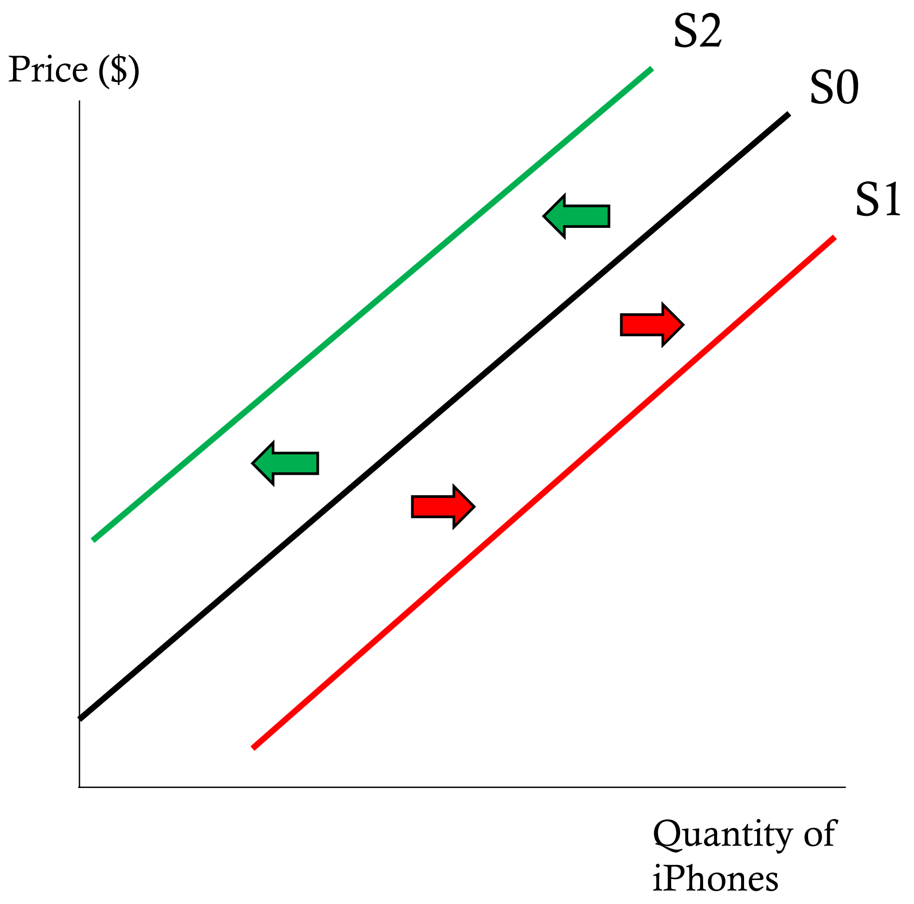
Before
Quantity = 40

After
Quantity = 30

What Shifts Supply?

- The ceteris paribus assumption (all things equal/held constant)
- Remove that assumption to allow factors to affect shift
- Factors (determinants)
  - Producers
  - Technology
  - Price Expectations
  - Prices of Relevant Resources
  - Prices of Related Goods
  - Taxes and Subsidies

Number of Producers



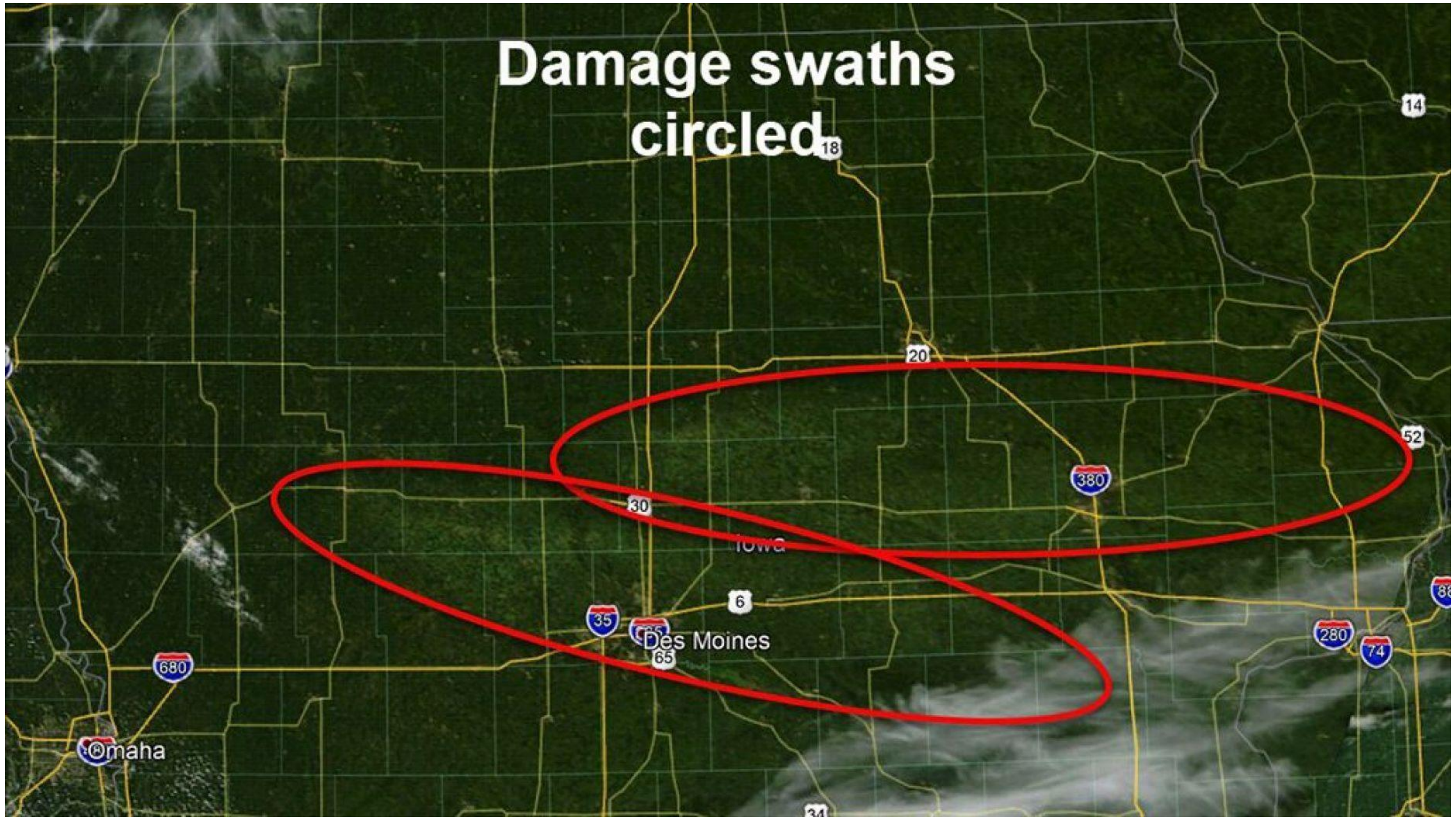
The number of producers directly affects the level of market supply.

If potential firms enter the market then supply increases

If firms leave the market then supply decreases

Number of Producers – Negative Shock

- [Derecho in Iowa Wipes out 10 Million Acres]
- 43 percent of Iowa's corn and soybean crop has been damaged or destroyed



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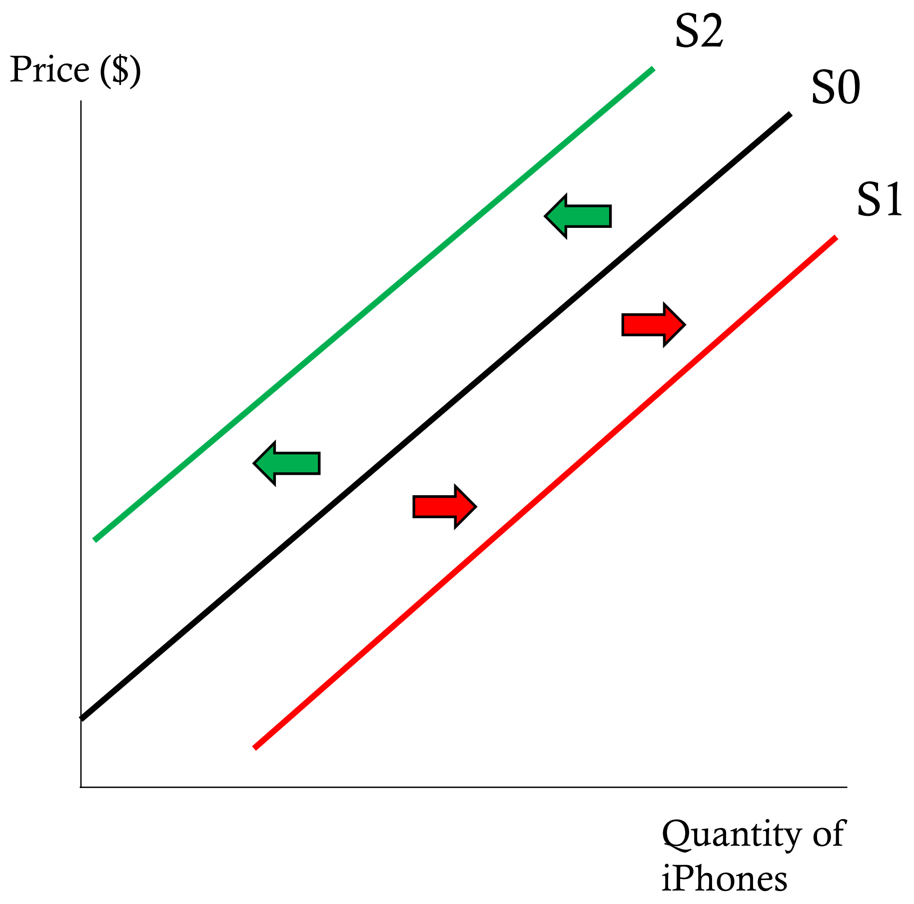
Technology



Recall: Technology is a way to make the production process more efficient.

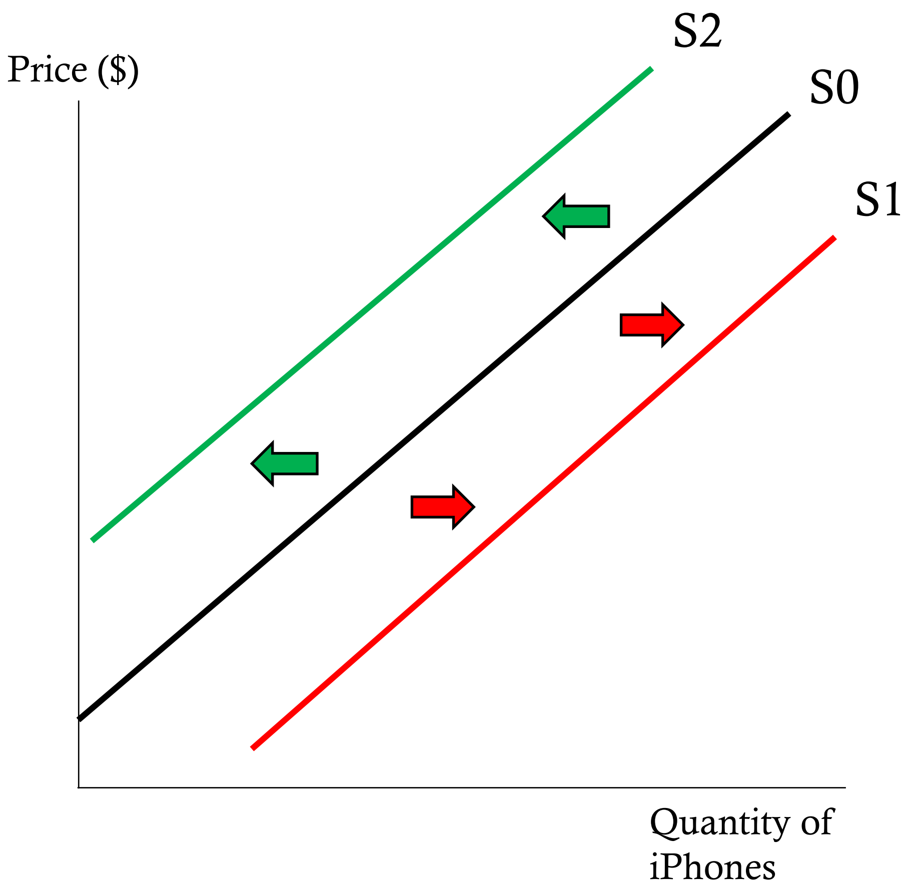
An increase in technology increases supply. This makes production cheaper and allows producers to increase quantity supplied at each previous price

A decrease in technology decreases supply. This makes production more expensive, and producers decrease quantity supplied at each previous price

Technology – What Hath God Wrought?

- Samuel Morse develops the telegraph in 1838
- The first message, “What hath God wrought?” traveled from Washington DC to a railroad depot in Baltimore in 1844.
- How does the invention of the telegraph help supply?
- More efficient means of communication between the different levels of suppliers of raw materials and firms
- In 1867, messages handled were 5.8 million with 85,000 miles of wire
- In 1877, messages handled were 21.1 million with 194,000 miles of wire
- By 1887, messages handled had increased to 47.4 million with 525,000 miles of wire
- The rise of the telephone would be the downfall of the telegraph

Price Expectations



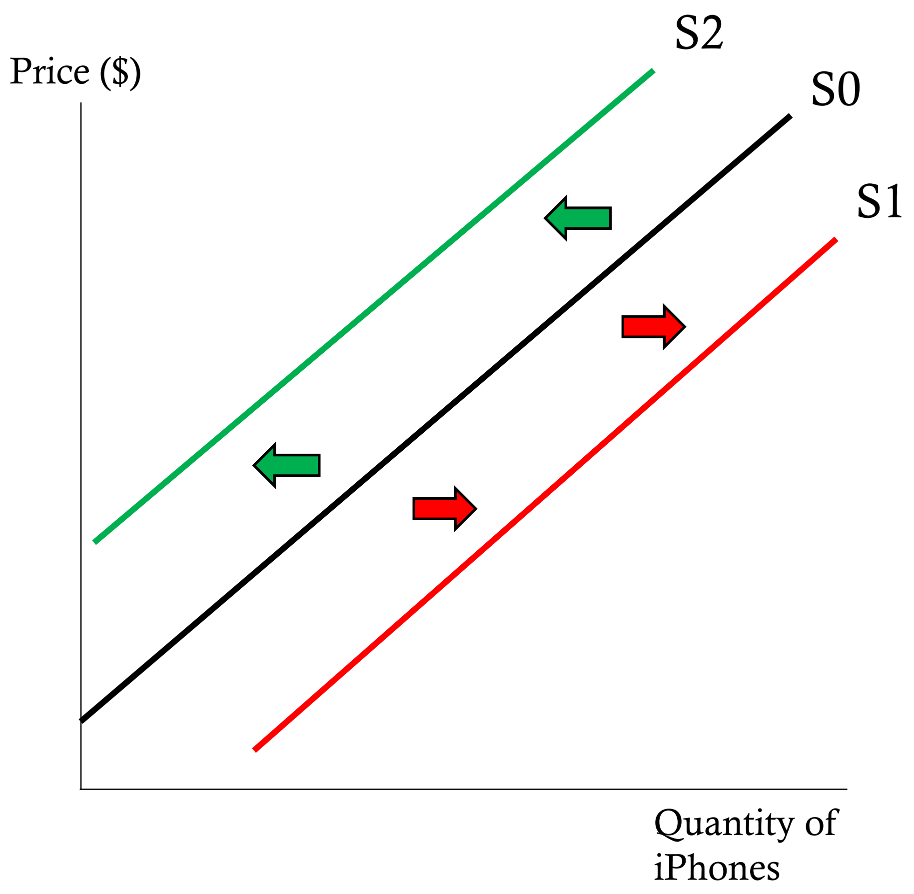
Like consumers, producers also have price expectations.

However, producers prefer higher prices as they can lead to greater profit.

A decrease in the expected future price increases supply today.

An increase in the expected future price decreases supply today.

Prices of Relevant Resources



Relevant resources are those resources used in the production process. For the production of iPhones, wheat would not be a relevant resource

A decrease in the price of a relevant resource increases supply.

An increase in the price of a relevant resource decreases supply.

Relevant Resources – A Sobering Effect

- Coronavirus to Burst Another Bubble? Carbonization Shortage Threatens Seltzer, Beer and Soda
- "Soft drink and beer makers are scrambling for access to CO2, a key ingredient in the carbonization used to make their products after coronavirus shutdowns have closed off their access to the chemical."
- "A coalition of associations from beer and meat packers, which use it for slaughter and refrigeration, sent a letter to Vice President Mike Pence on April 7, 2020, expressing "strong concern" about the shortage and asking for government intervention as ethanol plants were forced to close in droves due to the coronavirus crisis. CO2 production at ethanol plants, which produce carbon dioxide as a byproduct from fuel production, was down about 20% at the time."

Prices of Related Goods (complements)

Related goods are goods that that could also be produced (complements in production) or produced instead of the good in question (substitutes in production).

Complements in Production – goods that are produced together usually as a bi-product.

Example: Diamonds

When diamonds are cut for jewelry diamond dust is left over, which is applied to drill bits to make them stronger.

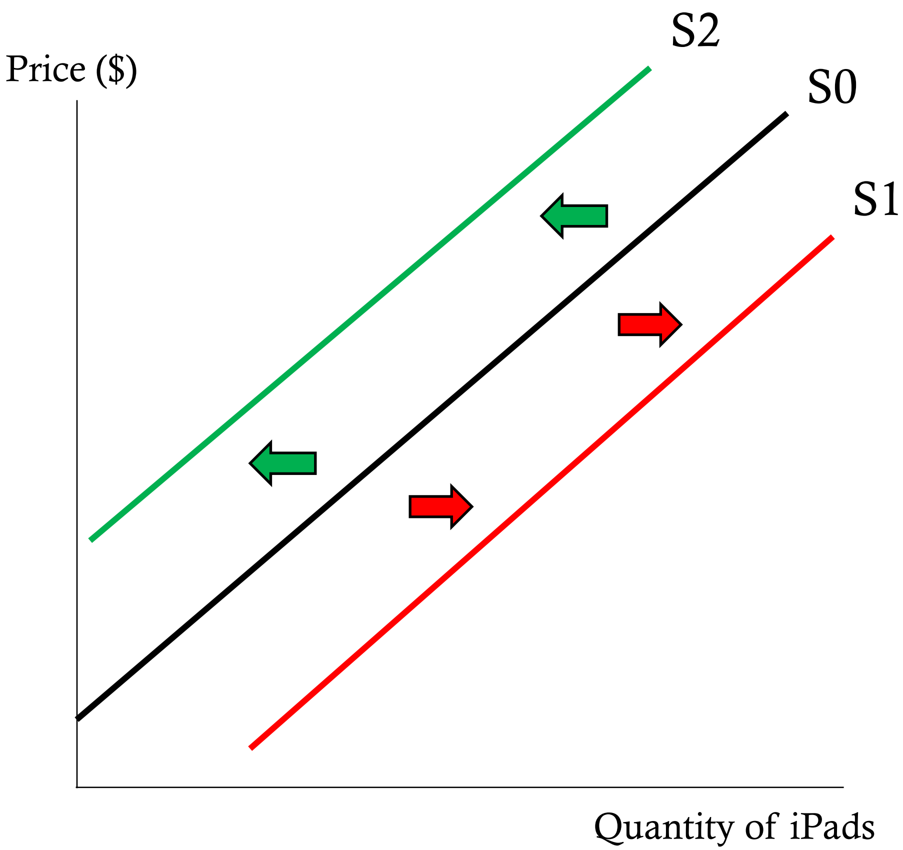
An increase in the price of diamonds increases the supply of diamond dust

A decrease in the price of diamonds decreases the supply of diamond dust

Related Goods – Masking Up

- 3M CEO: 'We're going 24/7' to ramp up production of masks to meet coronavirus demand
- "The CEO of industrial giant 3M said Tuesday that the company is increasing production of its respiratory protection products, such as masks, to meet demand caused by the outbreak of coronavirus."
- "We are ramping to full production. We're going 24/7," Roman said. He added that the company is increasing production at its plants in China and other Asian countries, as well as in Europe in the United States."

Prices of Related Goods (substitutes)



Substitutes in Production – goods that are produced in opposition of each other.

Example: iPads

The iPod touch and iPad may be viewed as substitutes (rather weakly)

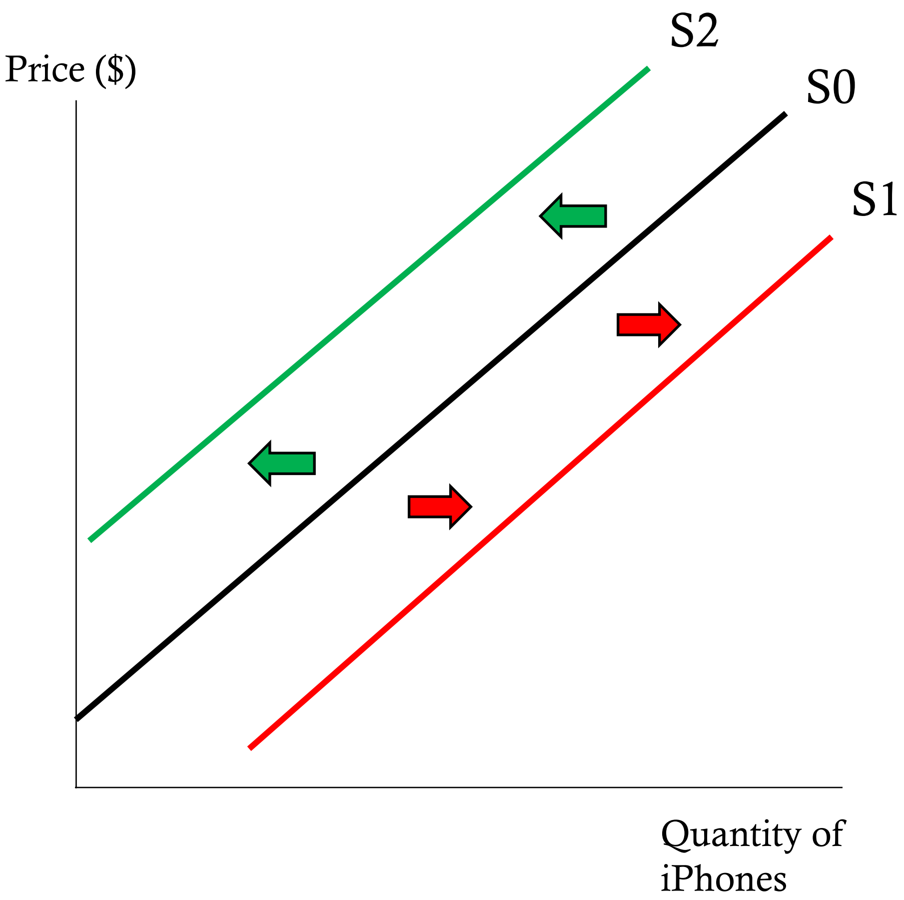
A decrease in the price of iPods increases the supply of iPads. This is because Apple would generate a greater profit by supplying more iPads than the now lower priced iPod.

An increase in the price of iPods decreases the supply of iPads, as Apple switches production to the now higher priced iPod.

Related Goods – Finding Other Things to Produce

- Tesla engineers are building ventilators for coronavirus patients out of car parts—take a look
- "A month ago, Tesla CEO Elon Musk expressed some skepticism about the severity of the coronavirus pandemic, but that hasn't stopped Tesla engineers from working to address the shortage of much-needed ventilators in American hospitals facing an influx of COVID-19 cases."
- "For instance, the schematic shows that the Tesla engineers are using parts typically found in the suspension system of a Tesla Model S, including the air tank and air-pressure regulator, to build a ventilator's oxygen mixing chamber. The engineers are also using the infotainment computer and touchscreen from a Model 3 to monitor and control a ventilator's air flow."

Taxes and Subsidies



Taxes and subsidies directly affect the cost of production by either making it more expensive (taxes) or less expensive (subsidies)

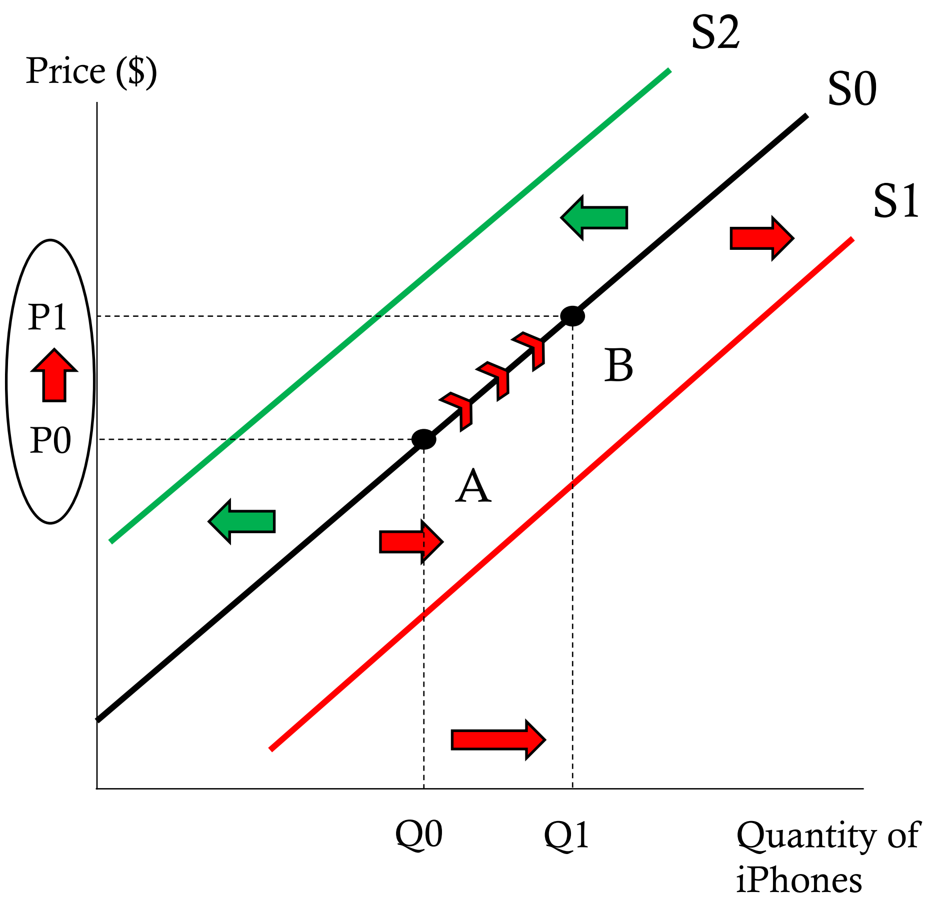
A(n) decrease/increase in taxes/subsidies will increase supply.

A(n) increase/decrease in taxes/subsidies will decrease supply.

Taxes and Subsidies – Tesla Cars

- Tesla cars built in China have been recommended for government subsidies, report says
- “China’s industry ministry has put Tesla Model 3 cars that are built inside the country on a list of vehicles recommended for government subsidies, according to a Reuters report on Friday.
- Reuters, citing a document published by the Ministry of Industry and Information Technology, said the level of subsidy that Tesla would receive was not yet clear. Two types of the Model 3 were on the recommendation list for new energy vehicle subsidies, it said.”

What Causes a Shift in Supply vs. Movement



Changes in quantity supplied are represented by movements along the supply curve (A to B).

The only thing that affects a change in quantity supplied is a change in price.

A change in supply is represented by a shift of the supply curve.

One of the six determinants of supply causes a change in supply.

Do the Laws of Supply and Demand Always Hold?

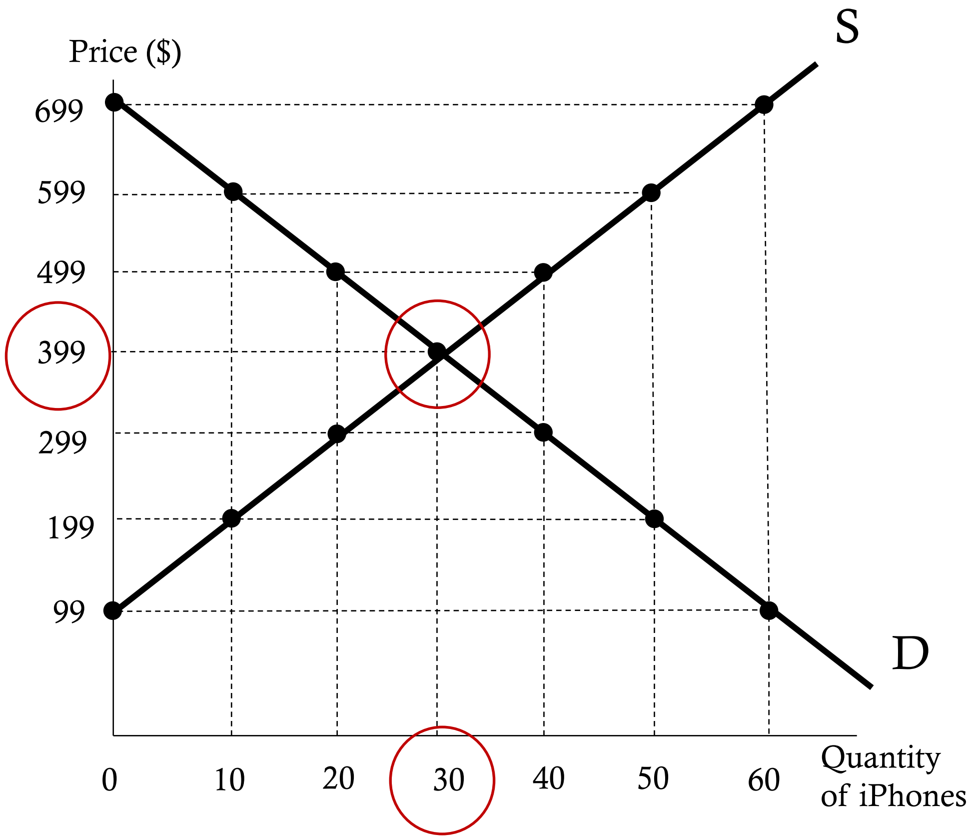
- We tend to act as these laws are iron-clad and further rises in price will always cause suppliers to put more to the market and a lower price will always bring more consumers to the market.
- However, these laws aren't iron-clad.
- Let's consider the market for blood donors.
  - Donating blood is typically seen as altruistic. However, there are many who may donate simply for money.
  - Yet, the perception that constantly donating blood for money may make some uneasy and hence any further rise in price will still prohibit them from donating.

Demand and Supply

- Where the demand and supply curves intersect, we say **market equilibrium** has been established. Below the market equilibrium is found at a price of $399 and 30 iPhones.

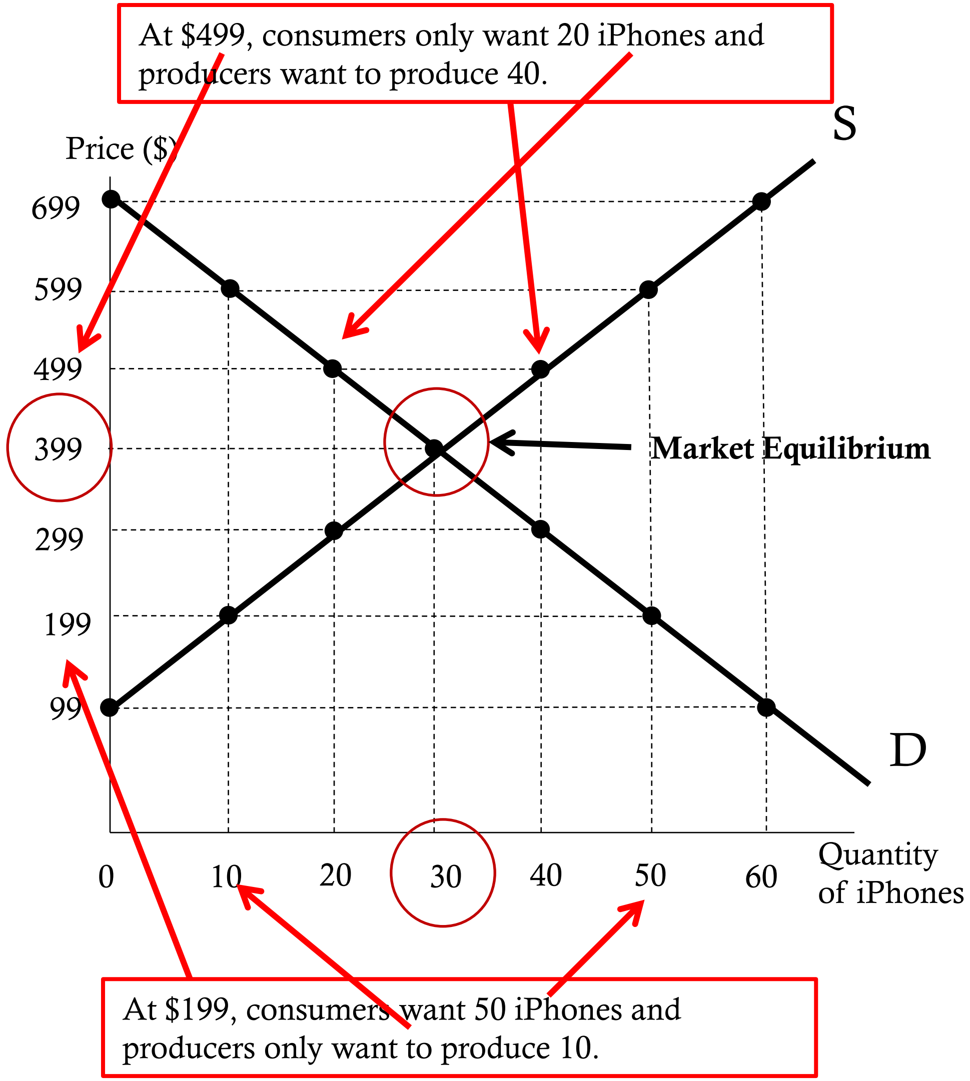
- Market equilibrium is also found in the demand and supply schedule where quantity demanded equals quantity supplied

Price	Quantity Demanded	Quantity Supplied
$699	0	60
599	10	50
499	20	40
399	30	30
299	40	20
199	50	10
99	60	0



Market Equilibrium

- Equilibrium is the condition where price regulates how much is bought and sold.
- Consumers want to buy 30 iPhones at a price of $399, and producers want to sell 30 iPhones at a price of $399.
- Any amount above the equilibrium price leaves consumers wanting less and producers wanting to sell more.
- Any amount below the equilibrium price leaves producers wanting to sell less and consumers wanting to buy more.



Excess Demand and Excess Supply

- The market is in equilibrium when quantity demanded equals quantity supplied. At a price of $399, we find market equilibrium because quantity demanded equals quantity supplied.

- If price is higher than $399 then the quantity supplied is greater than quantity demanded. This is called a **surplus**

- If price is lower than $399 then the quantity demanded is greater than quantity supplied. This is called a **shortage**

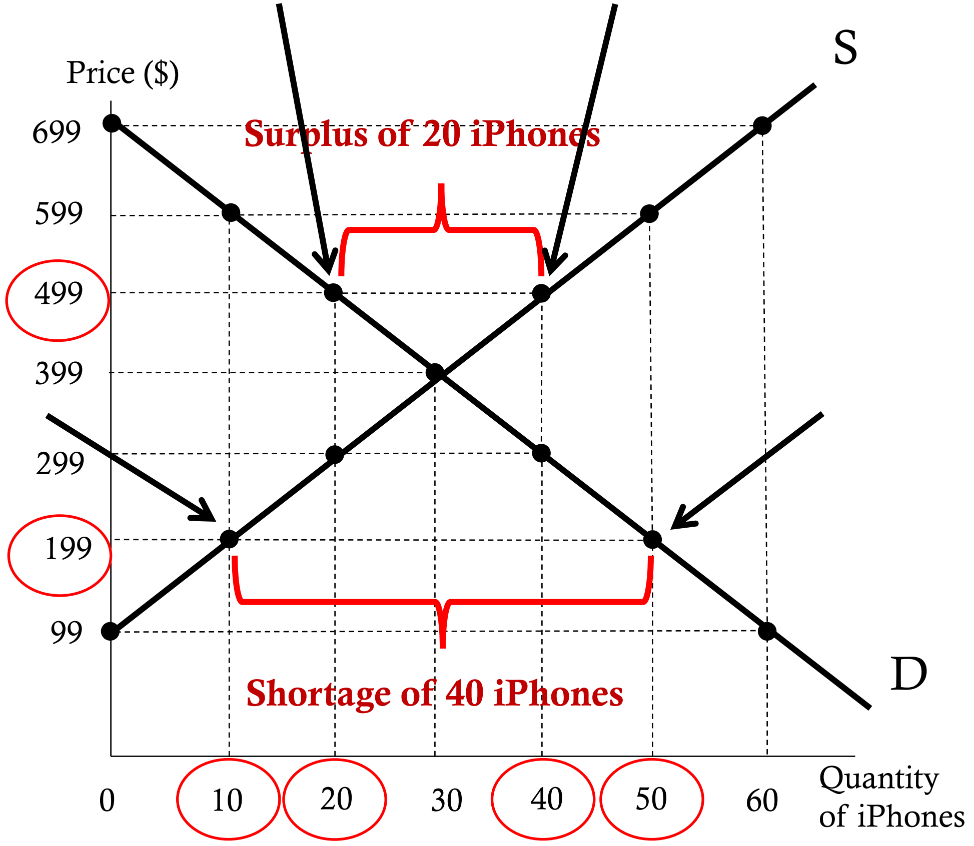
Price	**Quantity demanded**		**Quantity supplied**		**Surplus (-) Shortage (+)**
$699	0	—	60		- 60
599	10	—	50		- 40
499	20	—	40		- 20
399	30	—	30	=	0
299	40	—	20		20
199	50	—	10		40
99	60	—	0		60

- Remember the willingness and ability to pay or produce principle. At a price of $599, only 10 iPhones are bought. This means consumers were only willing and able to purchase 10 iPhones at a price of $599.

- At a price of $599 producers are willing to sell 50 iPhones but are only able to sell 10. These actions buy consumers and producers create the surplus of iPhones.

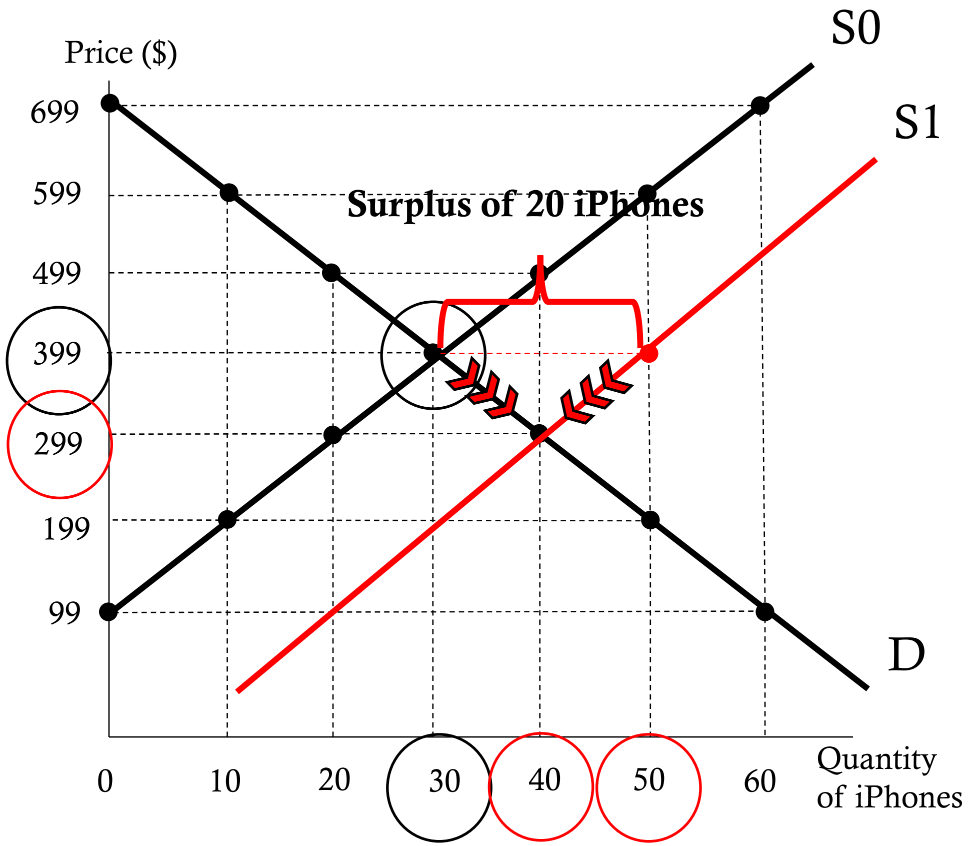
Surplus and Shortage

- Surpluses and shortages can be seen on the demand and supply graph as well.
- When the price is greater than equilibrium price a surplus occurs
- For example, at a price of $499 there is a surplus of 20 iPhones
- When the price is below the equilibrium price a shortage occurs
- For example, at a price of $199 there is a shortage of 40 iPhones
- **Note**: surpluses occur above market equilibrium, shortages occur below market equilibrium



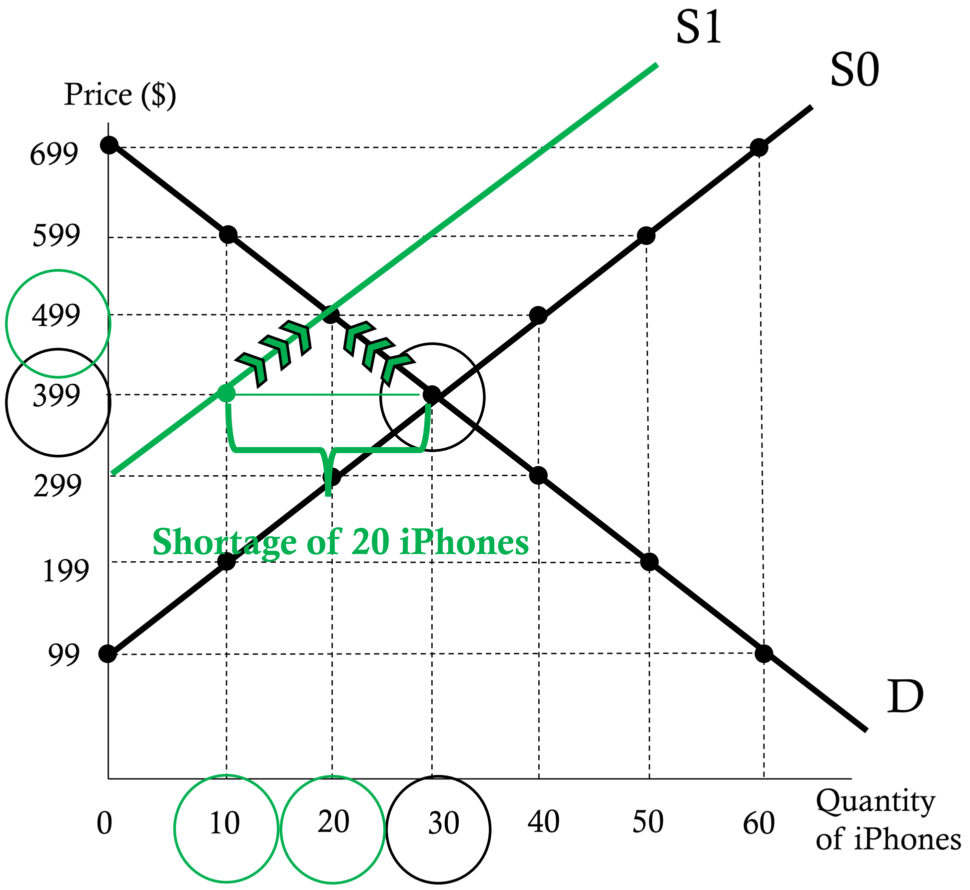
Surpluses, shortages, and shifts

- The market is in equilibrium at a price of $399 and 30 iPhones. There is no incentive to change for consumers or producers.
- Surpluses and shortages help us understand changes in market conditions.
- Suppose there is an increase in supply
- At the previous equilibrium price of $399 there is a surplus of 20 iPhones.
- To erase the surplus (or excess of iPhones) Apple will lower price.
- The new equilibrium price will be $299. As the price is lowered, consumers respond by purchasing more iPhones increasing quantity demanded to 40.
- **Recall**: a change in price causes a movement along the demand curve.



Surpluses, shortages, and shifts

- The original equilibrium price and quantity were $399 and 30.
- Suppose now we are interested in a decrease in supply.
- The supply curve shifts to the left and creates a shortage of 20 iPhones at the previous equilibrium price.
- Apple raises the price to erase the shortage.
- The new equilibrium price will be $499. Consumers respond by decreasing consumption and quantity demanded falls to 20.
- **Recall**: shortages are found below the market equilibrium. When the decrease in the supply curve occurred, the new equilibrium was found at the intersection of the demand curve and supply curve S1, which created the shortage.



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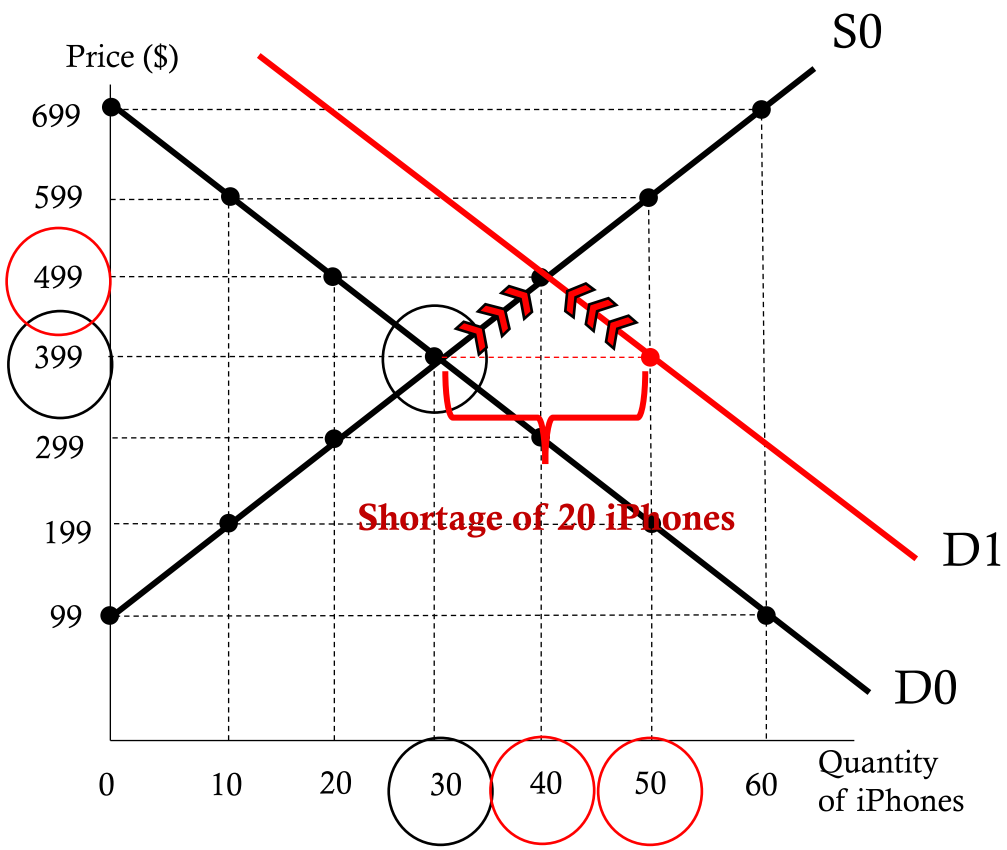
Surpluses, shortages, and shifts

- The original equilibrium price and quantity were $399 and 30.
- Suppose now we are interested in a increase in supply.
- The demand curve shifts to the right and creates a shortage of 20 iPhones at the previous equilibrium price.
- Apple raises the price to erase the shortage.
- The new equilibrium price will be $499. Consumers respond by decreasing consumption and quantity demanded falls to 20.
- **Recall**: shortages are found below the market equilibrium. When the decrease in the supply curve occurred, the new equilibrium was found at the intersection of the demand curve and supply curve S1, which created the shortage.



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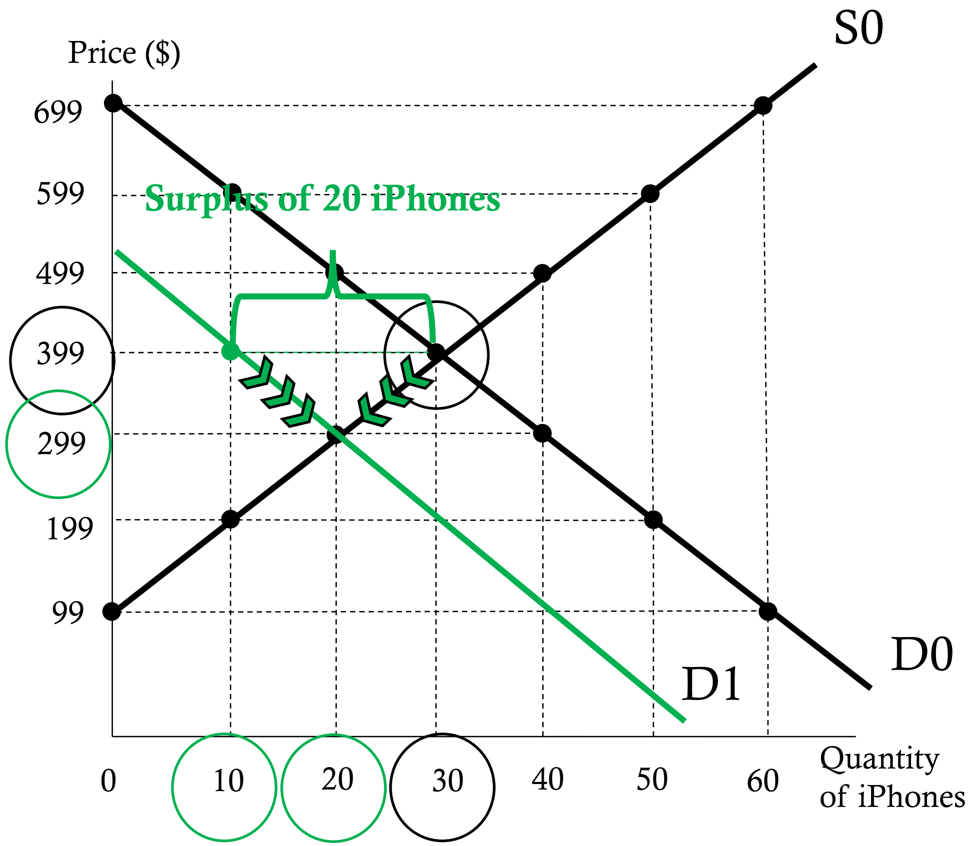
Surpluses, shortages, and shifts

- The original equilibrium price and quantity were $399 and 30.
- Suppose now we are interested in a decrease in demand.
- The demand curve shifts to the left and creates a surplus of 20 iPhones at the previous equilibrium price.
- Apple lowers the price to erase the surplus.
- The new equilibrium price will be $499. Consumers respond by decreasing consumption and quantity demanded falls to 20.
- **Recall**: surpluses are found above the market equilibrium. When the decrease in the demand curve occurred, the new equilibrium was found at the intersection of the demand curve and supply curve D1, which created the surplus.



Surpluses, shortages, and shifts

In each shift (demand or supply) changes in market equilibrium occurred. These changes affected both price and quantity.

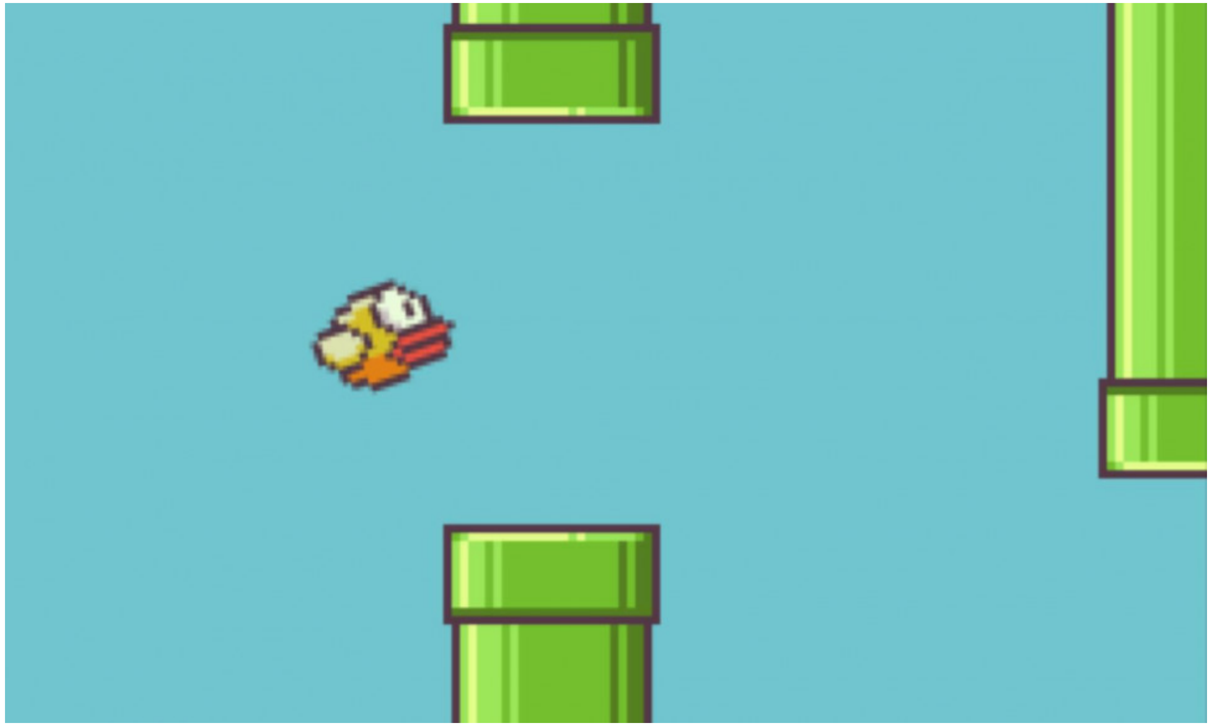
Demand	Supply
Increase	**Increase**
Price increased	Price decreased
Quantity increased	Quantity increased
Decrease	**Decrease**
Price decreased	Price increased
Quantity decreased	Quantity decreased

These results are important because they aid in the analysis if both curves shift as opposed to one curve shifting.

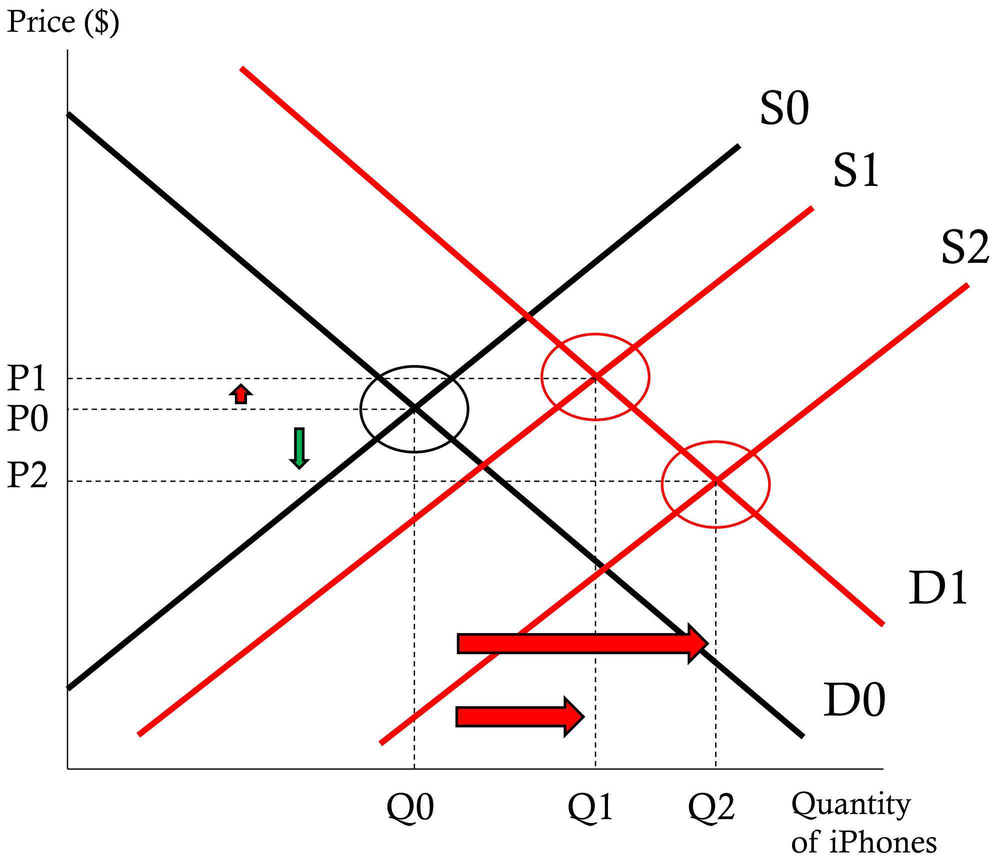
Suppose **demand** and **supply** both increased. The result is that quantity will increase as quantity increases for both, but the change in price is unclear as price increases with an increase in **demand** but decreases with an increase in **supply**.

Flappy Bird

“The highest bid for a phone offering "Flappy Bird" most recently sat at $14,900. A Sprint (S) iPhone 5 16GB reached that level after 26 bids and still has eight hours left on the auction. An iPhone 5s simply advertised as “Flappy installed bird,” sits at $8,100 with five bids.”

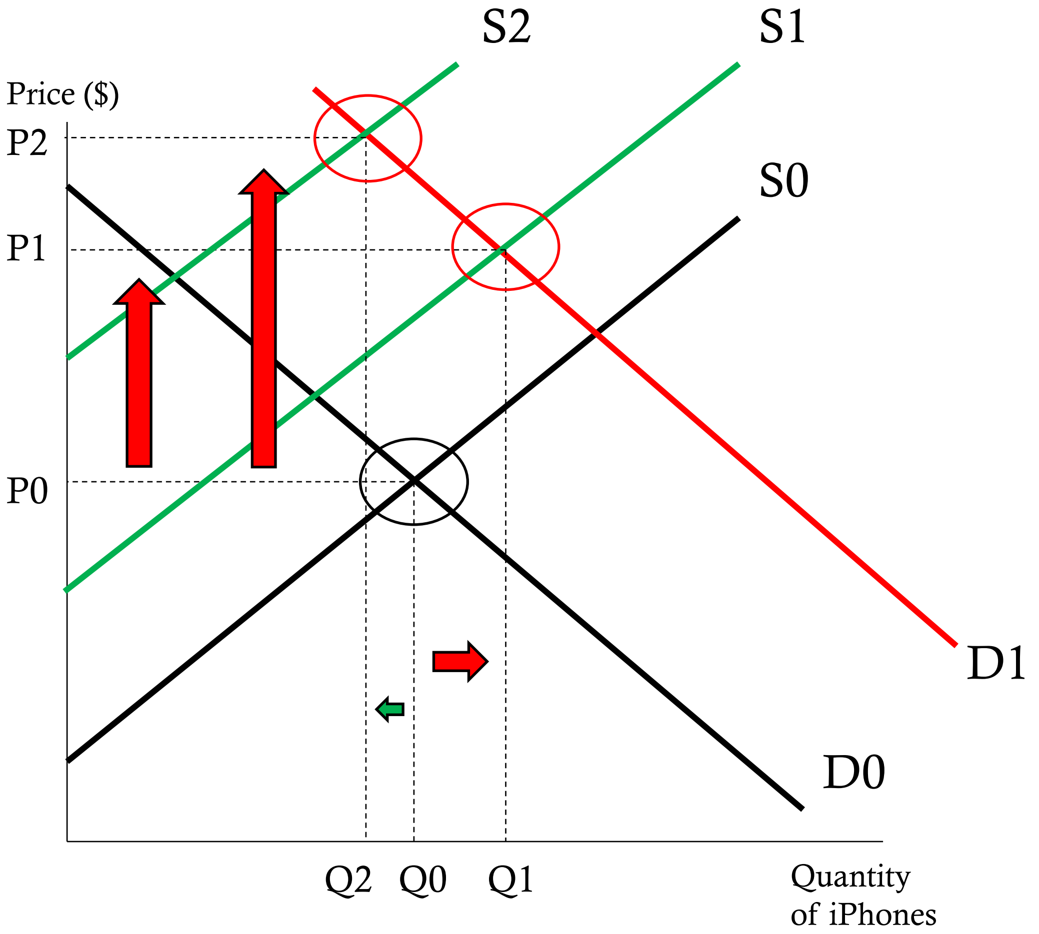


Simultaneous shifts in demand and supply: Increase in demand and supply



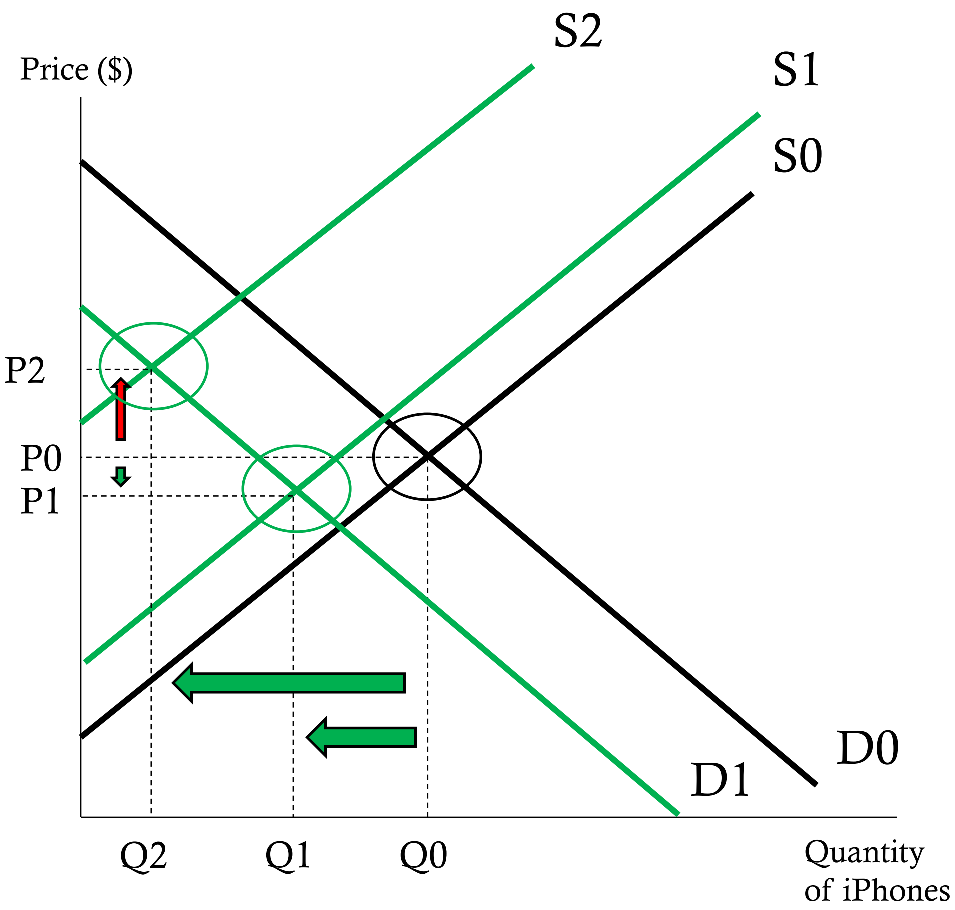
- Our market is in equilibrium at price, P0, and quantity, Q0.
- Suppose there is an increase in demand and supply, so that the new demand curve is D1 and the new supply curve is S1.
- At the new equilibrium, price is P1, and quantity is Q1.
- In this scenario, quantity increased as expected, and price increased as well.
- Suppose that supply increased to S2.
- Quantity increases to Q2, but now price falls below the original equilibrium price to P2
- Economist call price indeterminate because we don't know whether it will increase or decrease. It depends on the magnitude (how large or small) of the supply and demand curve shifts.

Simultaneous shifts in demand and supply: Increase in demand/decrease in supply



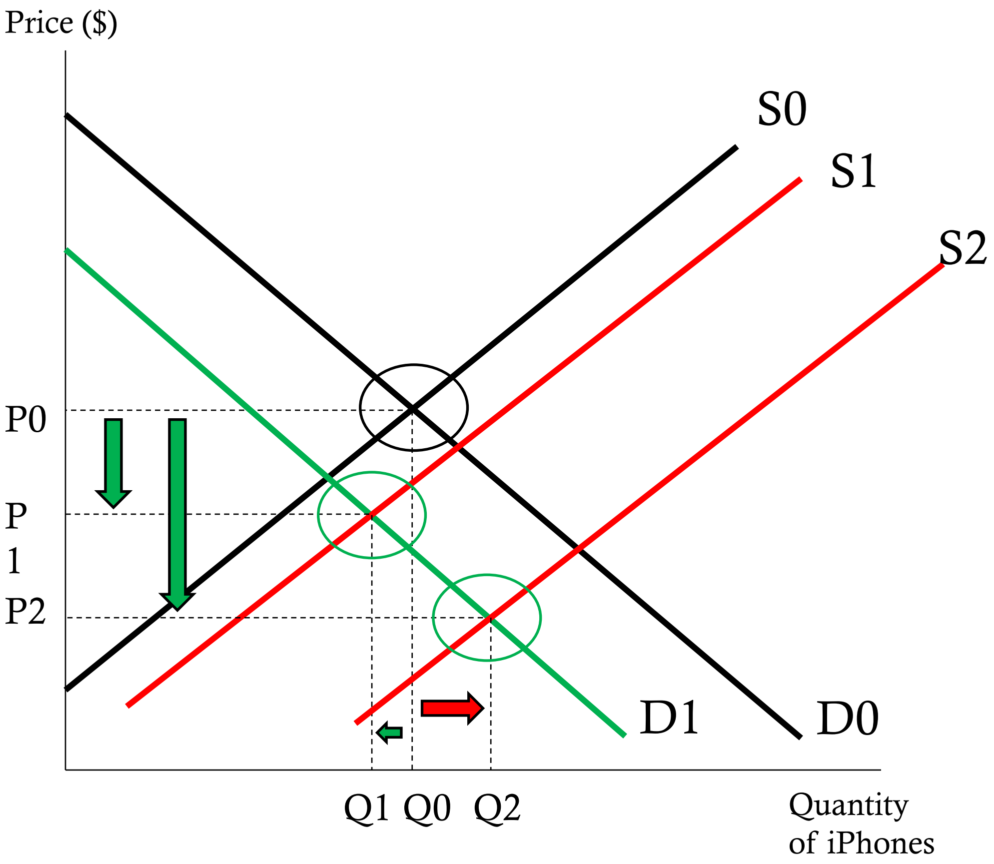
- Our market is in equilibrium at price, P0, and quantity, Q0.
- Suppose there is an increase in demand and a decrease in supply, so that the new demand curve is D1 and the new supply curve is S1.
- At the new equilibrium, price is P1, and quantity is Q1.
- In this scenario, quantity increased as expected, and price increased as well.
- Suppose that supply decreased to S2.
- Price increases to P2, but now quantity falls below the original equilibrium price to Q2
- Economist call price indeterminate because we don't know whether it will increase or decrease. It depends on the magnitude (how large or small) of the supply and demand curve shifts.

Simultaneous shifts in demand and supply: Decrease in demand and supply



- Our market is in equilibrium at price, P0, and quantity, Q0.
- Suppose there is a decrease in demand and supply, so that the new demand curve is D1 and the new supply curve is S1.
- At the new equilibrium, price is P1, and quantity is Q1.
- In this scenario, quantity decreased as expected, and price increased as well.
- Suppose that supply increased to S2.
- Quantity increases to Q2, but now price rises above the original equilibrium price to P2
- Economist call price indeterminate because we don’t know whether it will increase or decrease. It depends on the magnitude (how large or small) of the supply and demand curve shifts.

Simultaneous shifts in demand and supply: Decrease in demand/increase in supply



- Our market is in equilibrium at price, P0, and quantity, Q0.
- Suppose there is a decrease in demand and an increase in supply, so that the new demand curve is D1 and the new supply curve is S1.
- At the new equilibrium, price is P1, and quantity is Q1.
- In this scenario, price decreased as expected, and quantity decreased as well.
- Suppose that supply increased to S2.
- Quantity increases to Q2, but now price falls below the original equilibrium price to P2
- Economist call price indeterminate because we don't know whether it will increase or decrease. It depends on the magnitude (how large or small) of the supply and demand curve shifts.

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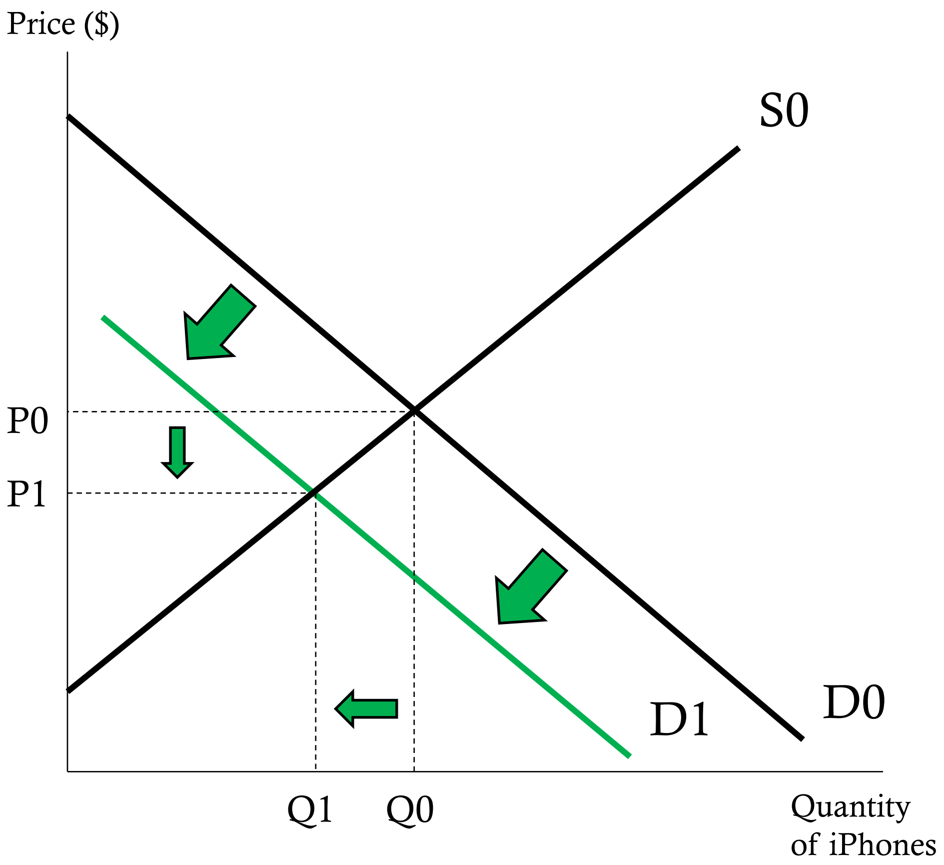
Simultaneous Shifts in Demand and Supply Summary

Demand	**Supply**	**Price**	**Quantity**
Increase	Increase	Indeterminate	Increases
Increase	Decrease	Increases	Indeterminate
Decrease	Decrease	Indeterminate	Decreases
Decrease	Increase	Decreases	Indeterminate

Example: Price of Related Goods

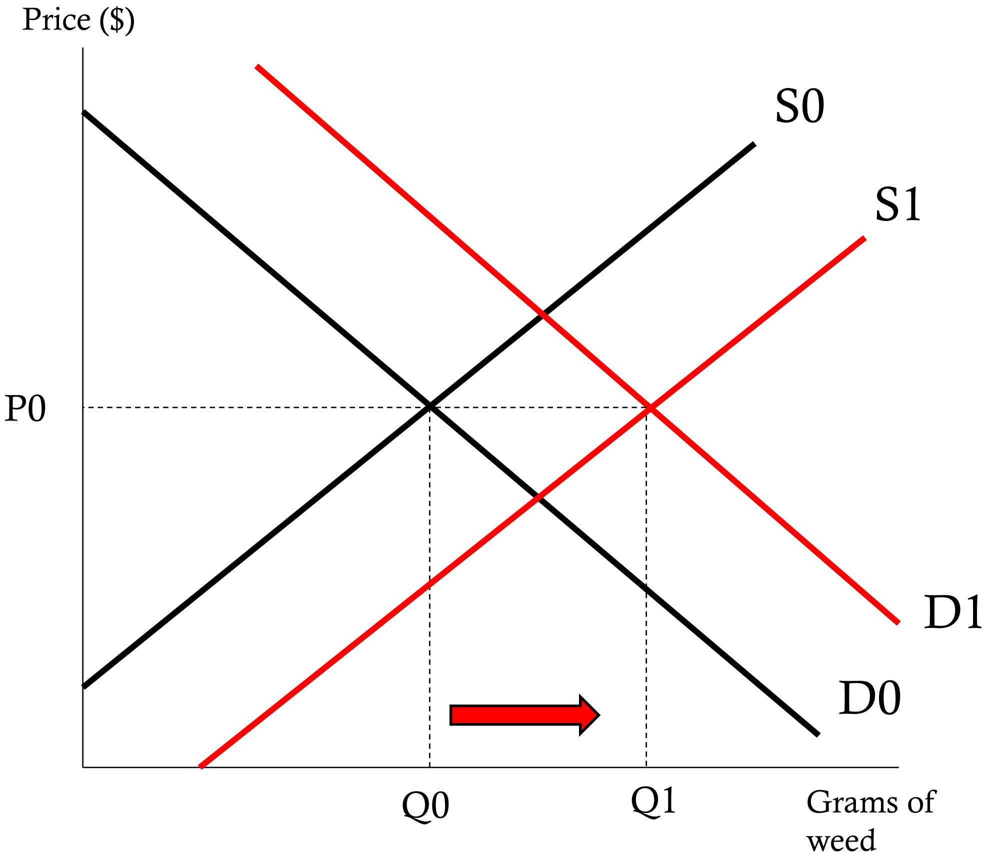


- Consider the market for iPhones.
- What is the result of a decrease in the price of the Samsung Galaxy A20?
- A decrease in the demand for Apple iPhones and a decrease in quantity supplied and price.
- The Galaxy A20 is a substitute for the iPhone. If the price of the A20 decreases, then the consumers substitute consumption for the iPhone for the A20. As a result, Apple provides less iPhones to the market.

Verizon

- "NEW YORK (Reuters) - Verizon Communications Inc's (VZ.N) quarterly revenue topped Wall Street analyst estimates on Thursday and the company added more phone subscribers than expected, sending shares of the No. 1 U.S. wireless carrier up in mid-morning trading."
- "Verizon said that it added 274,000 phone subscribers who pay a monthly bill on a net basis. Churn, or rate of defections for all customers paying a monthly bill, including tablet subscribers, was 0.97 percent."
- Of note, with the new iPhone 8 an X, many people who were own expired plans or soon to be expired plans with other carriers may have switched to Verizon to purchase the new phone.
- What do we expect to happen?

Marijuana Legalization - California



- What is the effect of a legalization of marijuana?
- “"The analysis estimated that as of November, aggregate annual sales in medical marijuana were $2 billion a year (about 25% of total marijuana sales) and sales in the illegal market were $5.7 billion (75%).”
- Demand and supply both increase. Quantity increases, but price is indeterminate.
- “According to the study, there will be increase in legal quantity demand, as recreational use increases, though no indication of what is going to happen to price. Of course, taxation will cause price to increase.”
- More people will begin smoking weed and more people will begin to sell weed.

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Key Takeaways

- **Markets coordinate buyers and sellers.**
  Competitive markets bring together many participants, so no single buyer or seller controls the price.
- **Demand shows willingness and ability to buy.**
  According to the law of demand, as price falls, quantity demanded rises. Determinants like income, tastes, population, expectations, and related goods shift demand.
- **Supply shows willingness and ability to sell.**
  According to the law of supply, as price rises, quantity supplied rises. Determinants like input prices, technology, producer expectations, related goods, taxes/subsidies, and the number of sellers shift supply.
- **Equilibrium price balances intentions.**
  The intersection of supply and demand sets the market price and quantity. At equilibrium, there's no shortage or surplus.
- **Prices signal and allocate resources.**
  When conditions change, prices adjust—encouraging buyers and sellers to alter behavior until a new equilibrium is reached.

Knowledge Check

1. If the price of iPhones decreases from $199 to $99, what happens to the quantity demanded of iPhones?
2. An increase in consumer income causes the demand for steak to rise. What type of good is steak?
3. A decrease in the price of computer chips reduces the cost of producing laptops. What happens to the supply of laptops?
4. If the market price of concert tickets is below equilibrium, what occurs?
5. When both supply and demand increase, what happens to equilibrium quantity?
6. What happens to equilibrium price and quantity if demand decreases and supply decreases?

Knowledge Check Answers

1. It increases. A lower price causes a movement *along* the demand curve, increasing quantity demanded.
2. Normal good. Demand for normal goods increases when income rises.
3. Supply increases. Lower input prices shift the supply curve to the right.
4. Shortage. At prices below equilibrium, quantity demanded exceeds quantity supplied.
5. It increases. Both shifts push quantity up, though the direction of price depends on which shift is larger.
6. Quantity falls. Price is indeterminate. Both shifts lower quantity, but the price effect depends on relative magnitudes.